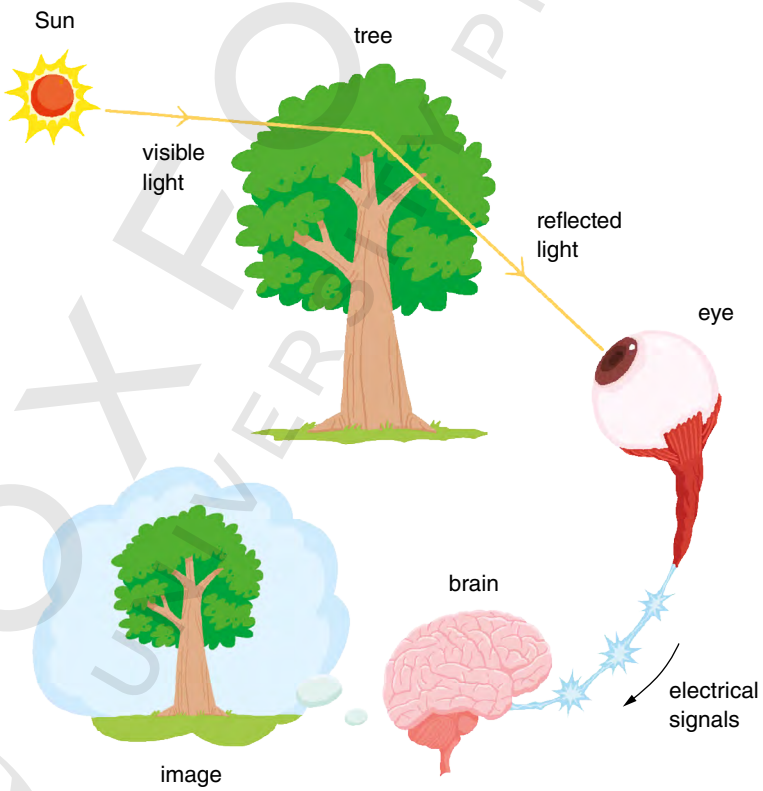
Introduction to medical physics

In this book, we will discuss how our knowledge of physics can be applied to medicine. Medical physics is a very wide field and we will focus on two specific areas.

a Perceiving the environment

The first area deals with how our bodies detect radiation (light or sound) and make use of it to perceive the environment around us. The following figure illustrates the general underlying principle of how our bodies detect and interpret the interaction between radiation and the environment, using a tree as an example.



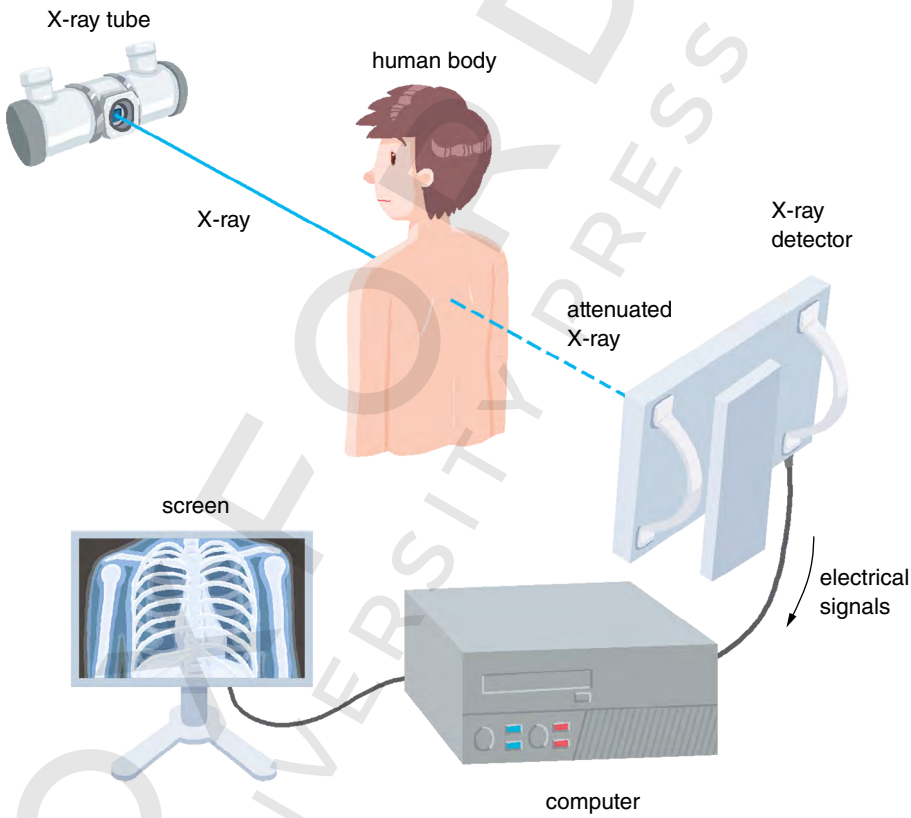
1. Radiation (visible light) is sent to an object (tree).
2. The radiation interacts with the object and is transformed (reflected) by this interaction.
3. The transformed radiation (reflected light) emerging from the object is detected by an organ (eye) and converted into electrical signals.
4. The signals are interpreted by the brain.
5. The brain makes sense of the object (recognizes an image of the tree).

Of course, in some cases, the detected radiation is emitted directly from an object, such as from a fire or a speaker.

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b Medical imaging

The second area that we will study is how medical instruments detect radiation and how we make use of them to investigate the organs inside a human body. The general underlying principle is exactly the same as in the previous case. This is illustrated in the following figure.



1. Radiation (X-ray) is sent into the body part which we want to investigate (chest).
2. The radiation interacts with the internal composition of the body part and is transformed (attenuated) by this interaction.
3. The transformed radiation (attenuated X-ray) emerging from the object is detected by an instrument (X-ray detector) and converted into electrical signals.
4. The signals are processed by another device (computer).
5. The device outputs the processing result to a display unit (screen).

Again, in some cases, the detected radiation is emitted directly from the body part to be investigated.

We will see that these instruments are very useful for finding out the causes of different diseases.



1

In this chapter, we will learn about

- the structure of the eye and how we see things
- defects of vision and their correction
- fibre optic endoscopy

Human Vision and Fibre Optic Endoscopy

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1.1 Human vision

The human eye is the sensory organ responsible for vision and is one of the most complex organs of the human body. It captures light and sends signals to the brain for interpretation. It is worth learning more about the human eye.

Flipped classroom

Watch a video and answer the questions.



Fig 1.1a *Our eyes provide us vision.*

In junior science, we learned the structure of the human eye and how we see things. Let's have a review.

Simulation

→ This simulation shows a 3D model of the eye. It is available on OUP web in PC version (.exe) only.

Pre-study The human eye

The figure below (Fig a) shows the basic parts of the human eye.

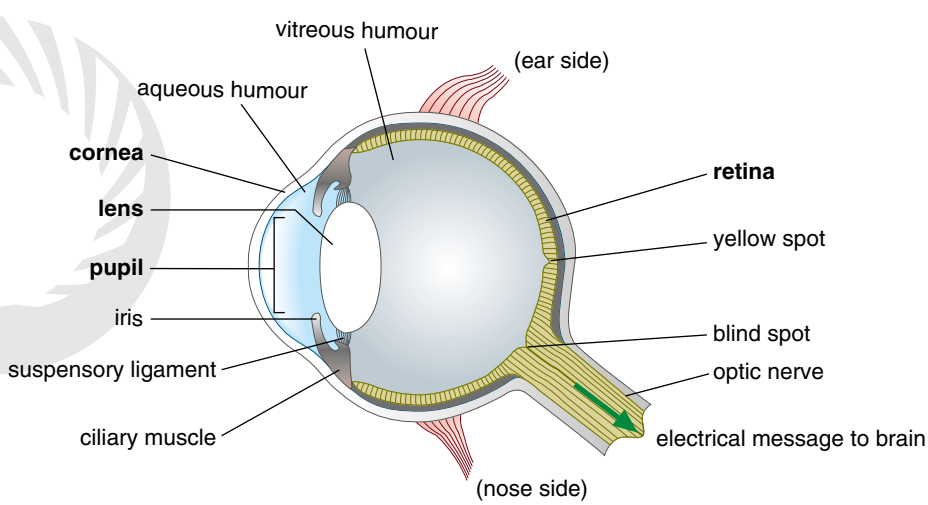


Fig a *A right eye as viewed from above.*

Some Ss may confuse iris and pupil. Iris is a circular sheet of muscle at the front of the eyeball whereas the pupil is an opening at the centre of the iris.

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When we look at an object, the incoming light from the object passes through the *pupil* to form an image on the *retina* (Fig b). It is refracted by the *cornea* and the *lens* in the process. The retina is sensitive to light and sends electrical signals along the *optic nerve* to the brain.

Note that the image formed on the retina is inverted, but is processed and interpreted as erect by the brain.

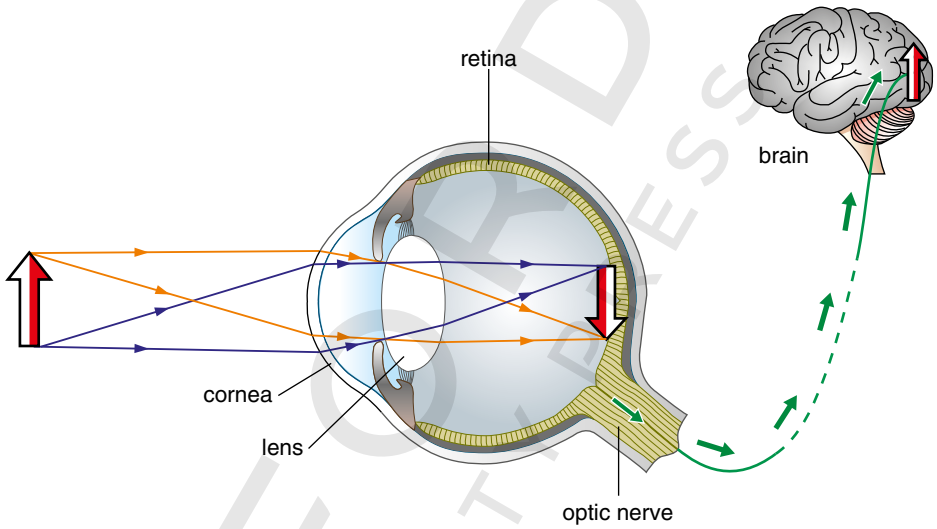


Fig b *Forming an image on the retina.*

DSE goal
Describe the process of accommodation of the eye.

1 Accommodation

In the eye, the cornea and the lens are the **refracting parts**. They act as a convex lens to refract light and form an image on the retina. Let us have a review of what we have learned in Book 3A Chapter 3 first.

a Review of lens and the lens formula

A convex lens can converge light rays while a concave lens can diverge light rays (Fig 1.1b).

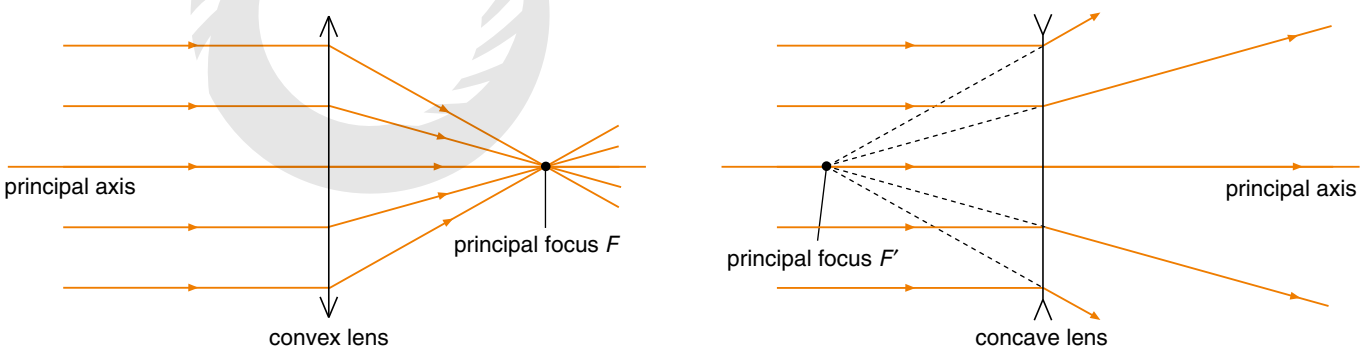


Fig 1.1b *Convex lens and concave lens.*

pupil 瞳孔 retina 視網膜 cornea 角膜 lens 晶體 optic nerve 視神經

When an object is placed in front of a lens, its image can be formed at different positions, depending on the focal length of the lens and the distance of the object from the lens. The object distance u, image distance v and the focal length f of the lens are related by the lens formula (Fig 1.1c):

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

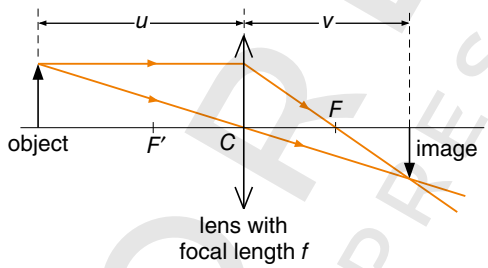


Fig 1.1c *u, v and f (using convex lens as an example).*

When using the formula, the 'real-is-positive' convention is adopted (Table 1.1a).

✎ Remind Ss what real and virtual images are.

	f	*u*	*v*	
			Real image	**Virtual image**
Sign	+ (convex lens)	+	+	–
	– (concave lens)	+	impossible	–

Table 1.1a *Sign of u, v and f.*

b Changing the shape of lens

If we want to see an object clearly, its image should be formed on the retina. Otherwise, the object would look blurred. Figure 1.1d shows how an image is formed by the refracting parts on the retina.

✎ Refraction allows large objects to be projected onto the limited area of the retina, i.e. miniaturize the image.

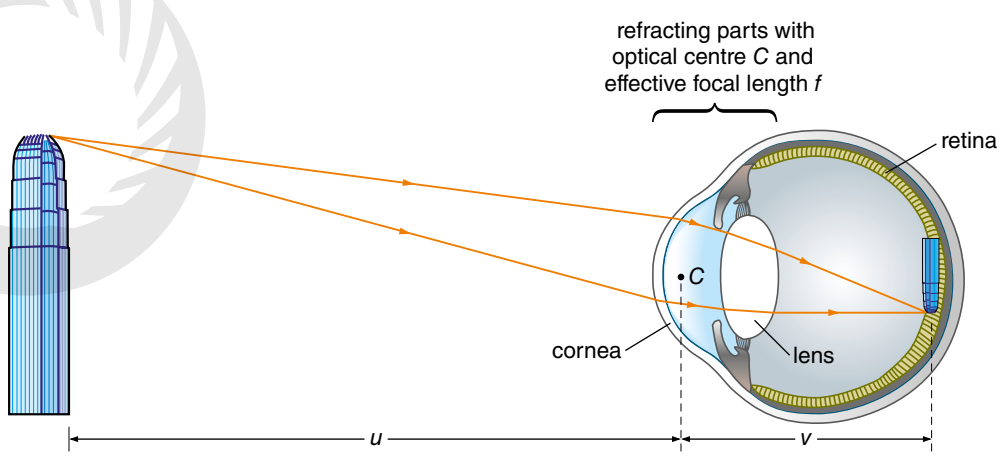


Fig 1.1d *Forming an image on the retina.*

The eyeballs have a rather constant shape and size. This means that the image distance v is fixed. When we want to see things at different object distances u clearly, we need to change the effective focal length f of the refracting parts to form the image on the retina. This can be done by changing the shape of the lens, which is elastic. This process of adjusting the lens is called **accommodation**.

Accommodation is the process of changing the shape (or focal length) of the lens to form a sharp image on the retina.

The cornea is fixed in thickness and shape, so only the lens can change the focal length (refracting power) for accommodation.

On the other hand, although the light is refracted the most at the surface of the cornea, this refraction is fixed and cannot be adjusted.

Know more

Refracting parts in the eye

A light ray is refracted when it passes between two media with different refractive indices. When a light ray enters the eye, it actually passes through several media and therefore is refracted several times (Fig a).

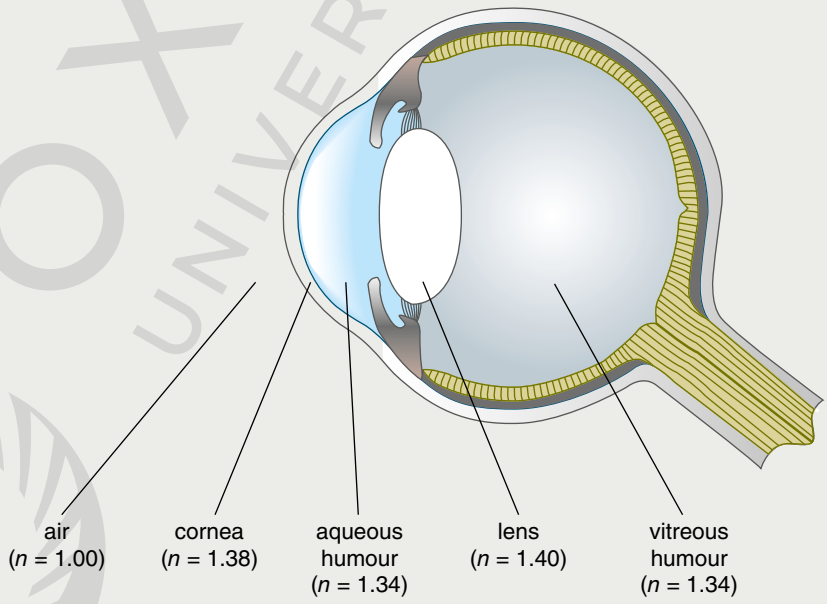


Fig a *Refractive indices n of different media.*

Note that:

1. The light ray is refracted most when it passes from air (n = 1.00) to the cornea (n = 1.38).
2. The refraction by the lens can be adjusted.

accommodation 視覺調節

The shape of the lens is controlled by the *ciliary muscles*, which hold the lens through *suspensory ligaments*.

When **looking at a near object** (small u), the **ciliary muscles contract** to make the **lens become thick** (small f) consequently. As a result, the light from the object is **refracted more** and can converge on the retina (Fig 1.1e).

When **looking at a far object** (large u), the **ciliary muscles relax** to make the **lens become thin** (large f) consequently. As a result, the light from the object is **refracted less** and can converge on the retina (Fig 1.1f).

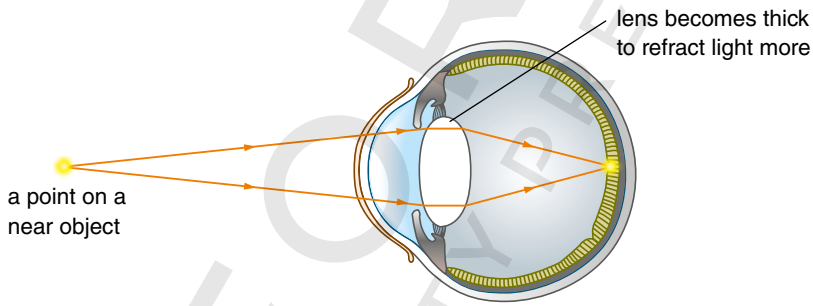


Fig 1.1e *Accommodation when looking at a near object.*

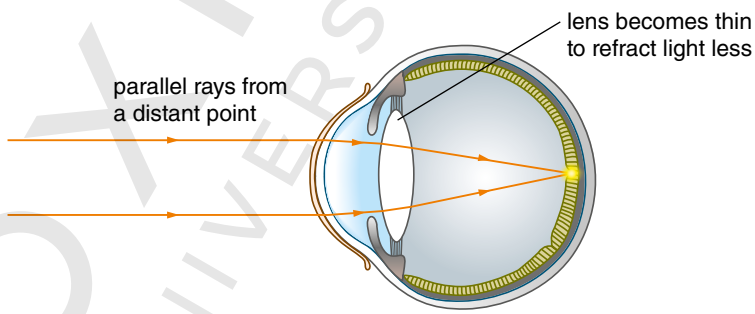


Fig 1.1f *Accommodation when looking at a far object.*

Know more

The ciliary muscles and the suspensory ligaments

Do you know how the ciliary muscles control the thickness of the lens?

▸ p.57

In simpler terms, light is bent more (refracted more) by the thicker lens.

✎ The need to change f in accommodation can be explained in another way:

Light rays from near objects are divergent while those from distant objects are parallel. These imply that light rays from near objects need to be bent more so that they can be focused on the retina. Since a more convex lens refracts light more, it is required to focus near objects; similarly, a less convex lens is required to focus distant objects.

Know more

Seeing objects at different distances at the same time

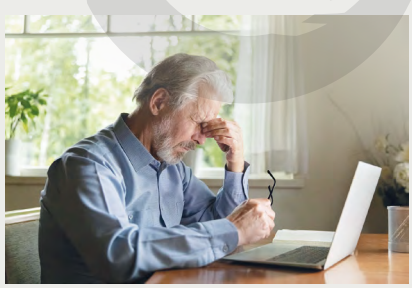
Can we see a near object and a far object clearly at the same time?

▸ p.57

Know more

Tiredness of ciliary muscles

If we look at near objects for a long time, the ciliary muscles will become very tired. It is a good practice to relax them by looking at faraway things for a few minutes after 30–40 minutes when watching screens or reading books.



Swimming googles

The refractive indices of the cornea and water are very close at 1.38 and 1.33 respectively. Therefore, light entering the eye through water cannot be sufficiently refracted. If a swimmer wears swimming googles, the air inside the googles allows better refraction and the image can then be properly formed on the retina. As a result, the swimmer can see clearly under water.



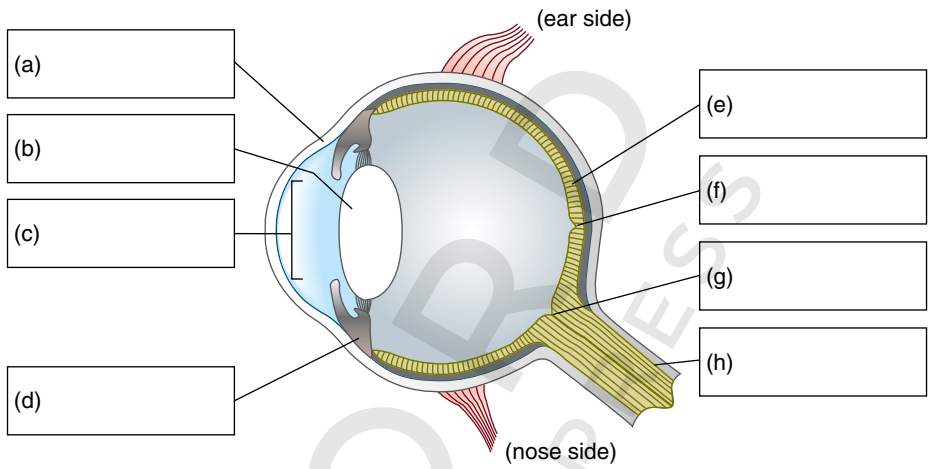
ciliary muscle 睫狀肌　　suspensory ligament 懸韌帶

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Complete the labels for the basic structures of the eye.



2 Which of the following are the major parts of the eye responsible for converging the incoming light? Put a tick in the corresponding box(es).

- ☐ Cornea
- ☐ Pupil
- ☐ Lens
- ☐ Retina

3 Which of the following are the major parts of the eye responsible for accommodation? Put a tick in the corresponding box(es).

- ☐ Cornea
- ☐ Pupil
- ☐ Ciliary muscles
- ☐ Lens

4 You are looking at this page. What will happen if you change to look at something outside through the window?

- The cornea refracts light (more / less / by the same degree).
- The lens refracts light (more / less / by the same degree).
- The lens becomes (more / less) convex or (thinner / thicker).
- The ciliary muscles (relax / contract).

5 You are staring at a mosquito. When will happen if it flies towards you?

- The cornea refracts light (more / less / by the same degree).
- The lens refracts light (more / less / by the same degree).
- The lens becomes (more / less) convex or (thinner / thicker).
- The ciliary muscles (relax / contract).

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Writing practice

Question 1

An object is moving away from the eye. Describe the accommodation process of the eye in order to keep seeing the object clearly. (2 marks)

Answer

The ciliary muscles in the eye relax 1A

to make the lens thinner. 1A

Question 2

By considering the lens formula, explain why making the lens thinner during accommodation can help an eye to see an object moving away from it clearly. (3 marks)

Answer

To see an object clearly, the image should be formed on the retina, i.e. the image distance v should be fixed. 1A

According to $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, as the object distance u increases, the focal length f has to be increased to keep v unchanged. 1A

During accommodation, the lens becomes thinner to increase its focal length f. 1A

Practise yourself

1 Sam first looks at a far object and then looks at a close object. Describe the accommodation process of his eye if he sees each object clearly. (2 marks)

2 Mandy is looking at the scene outside the window (Fig a). After a while, she looks at her computer and continues to work. By considering the lens formula, explain how the thickness of the lens of her eye is changed during accommodation. (4 marks)

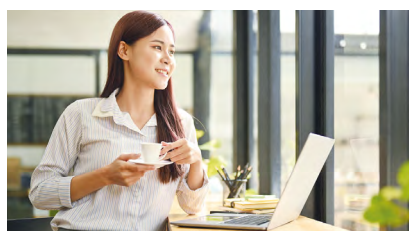


Fig a

2 Detection of light

Below we shall study what happens when light reaches the retina.

DSE goal

Describe the function of rods and cones.

DSE exam

16(2)Q4.1, 23(2)Q.4(a)

a Rods and cones

Light falling on the retina is received by the light sensitive cells in the retina. These cells can detect light and colour and act as **transducers**.

A transducer is a device which converts a form of energy signal (e.g. light or sound waves) into electrical signals or vice versa. The light sensitive cells in the retina convert different light waves/signals into nerve impulses.

There are two types of light sensitive cells which are responsible for our vision. They are **rod cells** (rods) and **cone cells** (cones).

Rods: They work best in low light conditions, and only provide black and white vision.

Cones: They are responsible for colour vision. They are more sensitive to changes in brightness than rods. However, they are only active at levels of high brightness (e.g. daylight). There are three types of cones most sensitive to different wavelengths of light.

See p.14 for further information.

Remember **C**ones for **C**olour.

Figure 1.1g shows the arrangement of rods and cones in the retina.

Rods and cones are named after the shape of their outer segment.

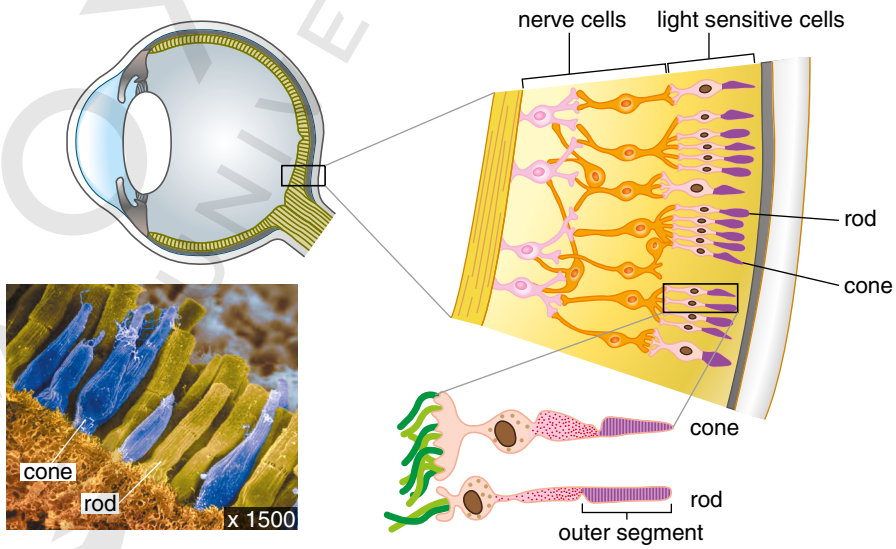


Fig 1.1g *Rods and cones in the retina.*

Rods are responsible for black and white vision and vision in low light conditions.

Cones are responsible for colour vision. They are only active at levels of high brightness.

transducer 換能器　　rod cell 視桿細胞　　cone cell 視錐細胞

The human retina contains many more rods (about 120 million) than cones (about 6 million), and these two kinds of light sensitive cells are not evenly distributed in the retina (Fig 1.1h).

✎ Because the yellow spot almost lies on the principal axis of the cornea and the lens, the image of an object directly in front of the eye will fall on this spot.

Most cones are concentrated at the *yellow spot*, which is near the centre of the retina. This high concentration of cones gives us the ability to distinguish very fine detail. When an object is in front of us, its image falls on this spot when we look at the object directly. Therefore, we can examine objects directly in front of us in the finest detail.

The Hands-on activity below shows the existence of blind spot. ▶

✎ We usually do not notice the blind spot as the blind spot effect of one eye is cancelled out by the normal view from the other.

Rods are distributed throughout the retina except at the yellow spot and the *blind spot*. The blind spot is a position which does not contain any light sensitive cell and is where the optic nerve leaves the eye. Any light falling on it cannot initiate nerve impulses to the brain. As a result, an image formed there cannot be seen.

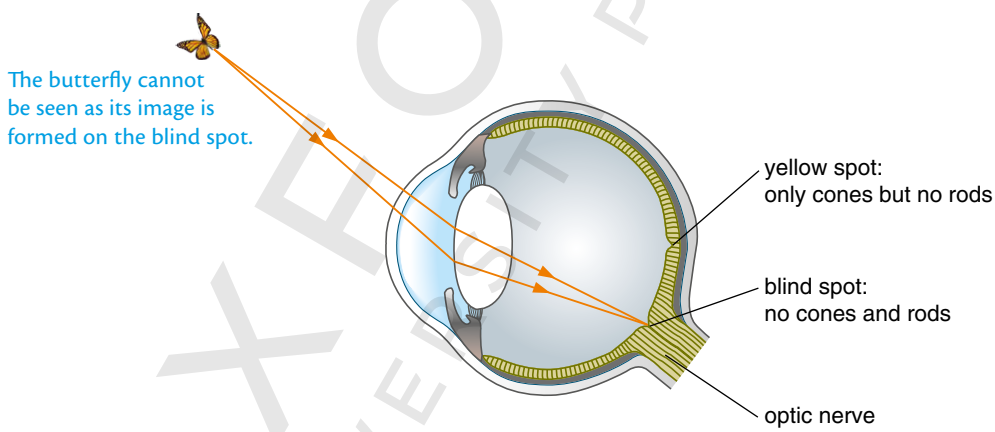


Fig 1.1h *Yellow spot and blind spot.*

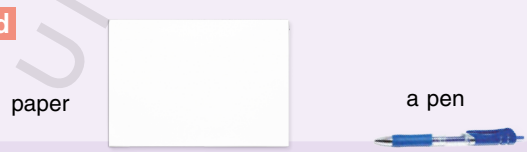
Video

→ This video demonstrates the Hands-on activity.

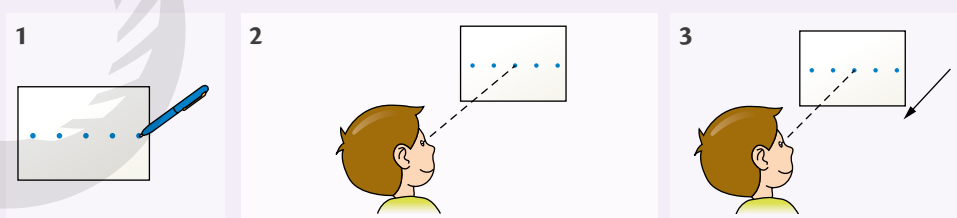
Hands-on activity

Finding your blind spot

What you need



What to do



What you find

What happens to the dots? Why does this happen?

yellow spot 黃點　　blind spot 盲點

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Complete the following table.

	Responsible for	Work well in
Rods	(colour / black and white) vision	(bright / dim) conditions
Cones	(colour / black and white) vision	(bright / dim) conditions

2 On the human retina, the number of cones is (much larger than / about the same as / much smaller than) the number of rods.

3 Which of the following will be the result if there is a deficiency or absence of rods in the eye?

A The colour of the objects cannot be distinguished.

B Objects cannot be seen under dim light conditions.

C No image can be formed on the retina.

4 Which of the following will be the result if there is a deficiency or absence of cones in the eye?

A The colour of the objects cannot be distinguished.

B Objects cannot be seen under dim light conditions.

C No image can be formed on the retina.

DSE goal

Interpret spectral response of light sensitive cells using receptor absorption curves.

b Spectral response and receptor absorption curves

Cones can detect light of wavelengths from about 400 nm to 700 nm. This is the range of wavelengths for visible light. The response of the cones to light of different wavelengths can be shown using **receptor absorption curves** (Fig 1.1i).

One can appreciate from the curves that each type of cones has a wide range of wavelength it can absorb light from. For example, green cones can also absorb light of 420 nm wavelength. They just are not as sensitive as blue cones at this wavelength.

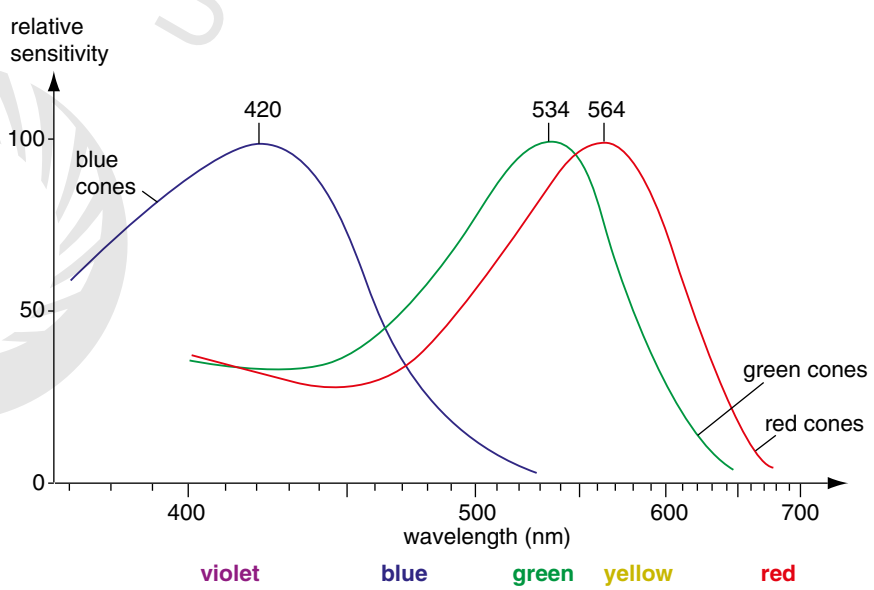


Fig 1.1i *Receptor absorption curves for cones.*

receptor absorption curve 感受器吸收曲線

From what colours the cones are sensitive to, it is no coincidence that: blue, green and red are defined as the three primary colours.

There are three types of cones which are most sensitive to red light (~564 nm), green light (~534 nm) and blue light (~420 nm), and are known as red cones, green cones and blue cones respectively.

When light of a certain wavelength reaches the retina, it stimulates different types of cones by different amounts. Then the cones send nerve impulses to the brain. Finally, the brain analyses the impulses and perceives the colour. The colour perceived depends on the combination of the stimulation strengths of the three types of cones. Figure 1.1j gives you a general idea on some of the colours perceived from different combinations of red, blue and green.

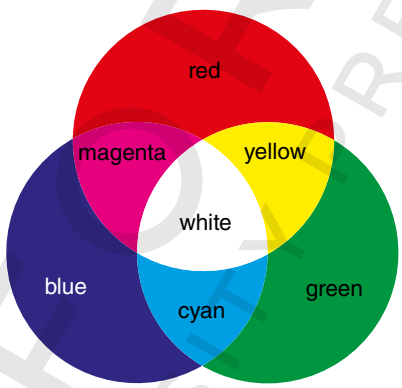
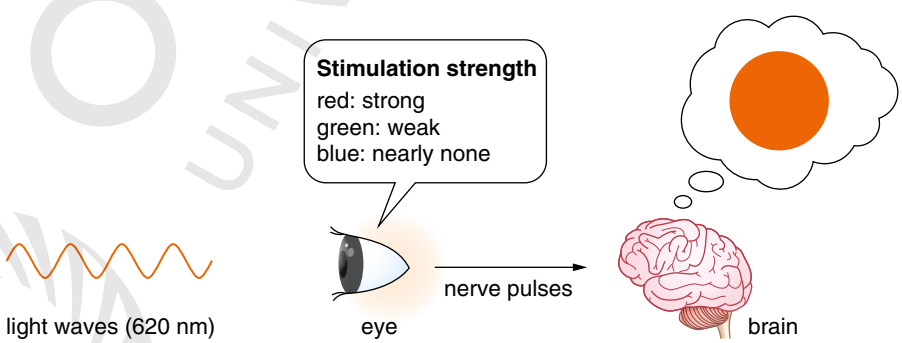


Fig 1.1j *Different combinations of red, blue and green give rise to some of the colours perceived.*

Figure 1.1k gives an example on how the brain perceives colour.



Rods are also stimulated by the light of this wavelength but they do not provide the colour vision.

A light of wavelength 620 nm causes significant stimulation to red cones, weak stimulation to green cones and nearly no stimulation to blue cones. As a result, the brain perceives the colour orange.

Fig 1.1k *How the brain perceives colour.*

On the other hand, the rods are most sensitive to light at about 500 nm (Fig 1.1l on p.15). As they cannot distinguish between colours, their response to light of different wavelengths is perceived as different shades of grey.

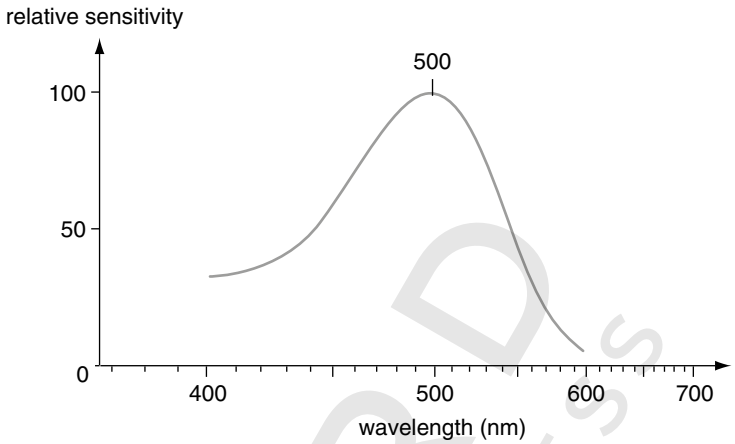


Fig 1.1l *Receptor absorption curve for rods.*

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Monochromatic light within the visible range falls onto the retina. Determine whether the following statements are true.

(a) Only one type of cones is stimulated. (T/F)

(b) The rods are not stimulated if the wavelength is not 500 nm. (T/F)

(c) Only greyscale images can be seen. (T/F)

(d) The colour perceived depends on the combination of the stimulation strengths of the three types of cones. (T/F)

2 Refer to the spectral response curves in Figure 1.1i on p.13. Determine which types of cones (red, green or blue) will be stimulated **the most** if monochromatic light of each of the following wavelengths falls onto the retina.

(a) 420 nm

(b) 500 nm

(c) 600 nm

DSE goal

Apply $\theta \approx \frac{1.22\lambda}{d}$ to solve problems.

DSE exam

13(2)Q4.2, 14(2)Q.4(b)(ii)&(iii)

3 Resolving power of the eye

If two objects are too close to each other, or if they are too far away from the eye, the eye will be unable to distinguish (resolve) them. The eye will see them as one object. For example, the two lights in Figure 1.1m will look like one light if they are gradually moved away from the eye.

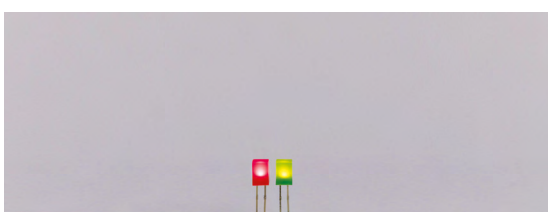


Fig 1.1m *What do you see when the two lights are far away from you?*

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Experiment 1a

Resolving power of the human eye (p.182) Video

→ This video demonstrates Expt 1a, in which the resolving power of the eyes is determined by looking at two closely-separated LEDs.

The **resolving power** of the eye is its ability to resolve the fine details of an object. The higher the resolving power, the finer the details the eye can resolve. What determines the resolving power of the eye?

a Angular separation of objects

The **angular separation** θ of two point objects is the angle subtended by the objects at the eye (Fig 1.1n).

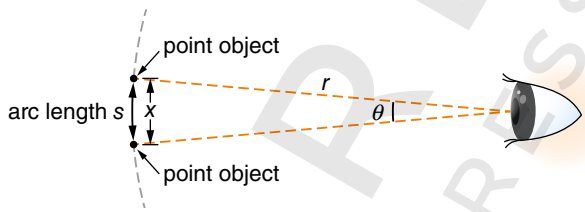


Fig 1.1n *Angular separation of two point objects.*

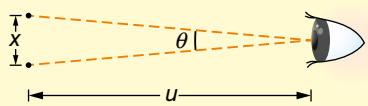
Since θ is usually very small, the separation x of the objects is nearly equal to the arc length s, i.e. $x \approx s$. Also, the distance of the objects from the eye u is nearly equal to r. If θ is in radian, $s = r\theta$, which implies that $x \approx u\theta$.

angular separation of objects (rad)

$$\theta \approx \frac{x}{u}$$

separation between objects (m)

distance between the objects and the eye (m)

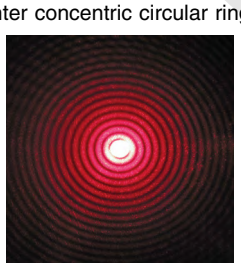


The minimum angular separation of two points at which they can be distinguished as separated by an eye is denoted by θ_R. In physics, diffraction of light imposes a theoretical limit on its value.

b Effect of diffraction on resolving power

How do λ and d affect diffraction?

When light passes through a small circular hole, it does not produce a bright dot on the screen behind the hole, but rather a circular disc known as Airy disc surrounded by much fainter concentric circular rings.



In Book 3B, we learned that light diffracts when it passes through a gap or a hole (such as the pupil of the eye). The degree of diffraction depends on the wavelength λ of light and the diameter d of the hole (Fig 1.1o).

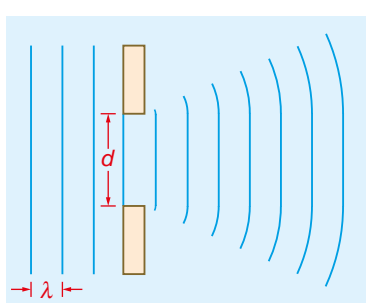


Fig 1.1o *Diffraction occurs when light passes through a hole.*

We mentioned that the light from a point object is converged to a point on the retina to form an image. However, due to diffraction as light enters the pupil, the image on the retina is actually a small disc.

resolving power 解像能力　　angular separation 角間距

Simulation

→ This simulation shows how the angular separation of two objects determines whether they can be resolved by the eye. It also shows how the resolving power of the eye is affected by the wavelength of light and the diameter of pupil. It is available on OUP web in PC version (.exe) only.

Consider an eye looking at two point objects (Fig 1.1p). When their angular separation gradually decreases, the diffraction patterns of their images become more and more overlapping until the eye is unable to resolve them.

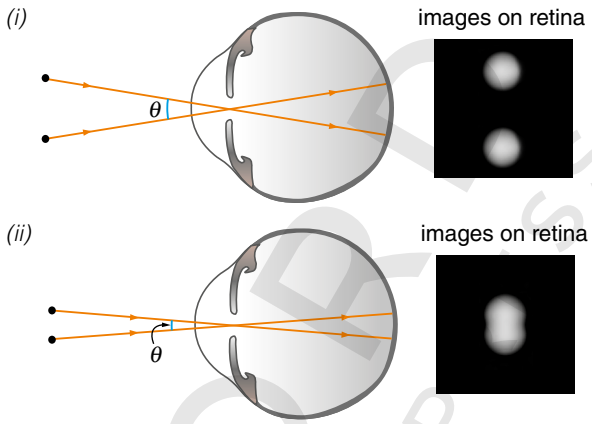


Fig 1.1p *(i) Resolvable objects ($\theta > \theta_R$), (ii) unresolved objects ($\theta < \theta_R$).*

The following relation gives the minimum angular separation θ_R of two points at which they can be distinguished as separated by the eye.

According to the Rayleigh criterion, two points can just be resolved when the central maximum of the diffraction pattern of one point is centred on the first-order minimum of the diffraction pattern of the other. It is expressed in a formula to find θ_R here. ▶

minimum **angular separation** of two resolvable points (rad)

$$\theta_R \approx \frac{1.22\lambda}{d}$$ **(Rayleigh criterion)**

wavelength of light (m)

diameter of pupil (m)

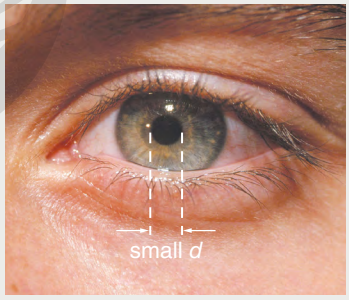
It is a measure of the resolving power of the eye. The smaller the value of θ_R, the higher the resolving power, and the finer the details the eye can resolve.

Know more

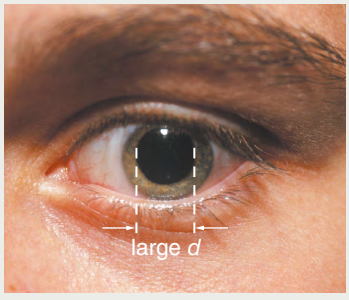
Size of pupil

See Fig a on p.4 to find out the positions of iris and pupil. ▶

The iris controls the size of the pupil and hence the amount of light entering the eye. The size of the pupil changes automatically depending on the surrounding light. It decreases in bright light and increases in dim light. The diameter of the pupil can typically change from 2 mm to 6 mm.



Pupil in bright light.

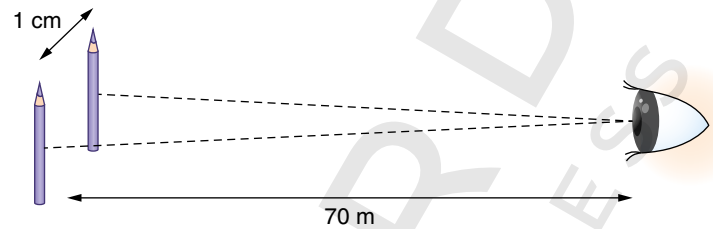


Pupil in dim light.

✎ Although our resolving power is higher in low light conditions, cones do not work well in these conditions. There is always a compromise between resolving power and contrast in our eyes.

Example 1 | Can two objects be resolved?

Sophia is observing, with only one eye, the tips of two purple pens which reflect light at 420 nm. The distance between the two tips is 1 cm and they are 70 m from her eye.



The diameter of the pupil of Sophia's eye is 3 mm.

(a) What is the angular separation of the two tips from Sophia's eye?

(b) Show that the two tips cannot be resolved by Sophia's eye.

(c) Describe how Sophia should change each of the following quantities to make the two tips resolvable.

(i) Distance between the two tips

(ii) Distance of the tips from Sophia's eye

Solution

(a) $\theta \approx \frac{x}{u} = \frac{0.01}{70}$

$= 1.43 \times 10^{-4}$ rad

The angular separation of the two tips from Sophia's eye is 1.43×10^{-4} rad.

(b) $\theta_R \approx \frac{1.22\lambda}{d} = \frac{1.22 \times (420 \times 10^{-9})}{3 \times 10^{-3}}$

$= 1.71 \times 10^{-4}$ rad

Since $\theta < \theta_R$, the two tips cannot be resolved by Sophia's eye.

Sophia should make θ larger. ▶

(c) (i) Sophia should increase the distance between the two tips.

(ii) Sophia should reduce the distance of the tips from her eye.

Think about...

Will the two tips be resolved more easily if the purple pens are replaced by red pens?

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Two points are resolvable by the eye when their angular separation θ is (larger than / equal to / smaller than) θ_R.

2 According to $\theta_R \approx \frac{1.22\lambda}{d}$,

- $\theta_R \uparrow$ when λ _____
- $\theta_R \uparrow$ when d _____

3 Complete the table.

	Wavelength of light λ	Diameter of pupil d	θ_R
(a)	6.32×10^{-7} m	3 mm	
(b)	450 nm	3.2 mm	
(c)	555 nm		1.9×10^{-4} rad

4 Complete the table.

	Separation of two points x	Distance of the points from the eye u	Angular separation of two points θ	Resolvable or not if $\theta_R = 1.8 \times 10^{-4}$ rad
(a)		50 m	1.95×10^{-4} rad	
(b)	2 cm		1.72×10^{-4} rad	
(c)	2.5 cm	36 m		

Learn by practice 1

1 An eye is looking at two point sources emitting orange light of wavelength 630 nm. The two sources are 0.025 m apart. The diameter of the pupil of the eye is 3.2 mm. Find

(a) the minimum angular separation of the two sources,

(b) the maximum distance of the two sources from the eye

so that they can be resolved.

2 Edward is observing two LEDs 90 m away. The LEDs are 1 cm apart and emit light of wavelength 450 nm. The diameter of the pupil of Edward's eye is 5 mm in this situation. Determine whether the two LEDs can be resolved by Edward's eye.

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Exercise 1.1

★**1** Which of the following is/are the change(s) during accommodation?

(1) The distance of the lens from the retina changes.

(2) The thickness of the lens changes.

(3) The amount of light entering the eye changes.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**2** Kim is looking at two orange objects at a distance but her eye cannot resolve them (Fig a).

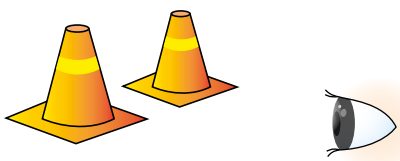


Fig a

Which of the following may allow her to resolve the two objects?

(1) Painting the objects blue

(2) Painting the objects red

(3) Walking towards the objects

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

3 Compare the two types of light sensitive cells on the retina in terms of their locations and functions.

4 An eye is looking at two points which reflect monochromatic light. θ_R is the minimum angular separation of the two points so that they can be resolved. Determine how the value of θ_R will change if each of the following changes is made.

(a) Light of longer wavelength is reflected.

(b) The diameter of the pupil decreases.

★**5** Sam is looking at two points which reflect green light of 555 nm. The two points are 80 m from Sam's eye. The diameter of the pupil of his eye is 3 mm.

(a) What is the minimum separation of the two points so that they can be resolved?

(b) How will the result in (a) change if the points reflect blue light instead of green light? Show your reasoning.

★**6** Figure b shows a cross-section of the human eye. Name structures *A*, *B* and *C* and state their functions.

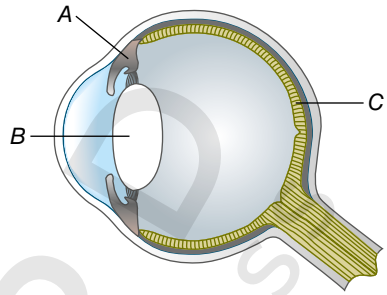


Fig b

★**7** Figure c shows the receptor absorption curve for green cones. Its peak is at 534 nm. What does this tell you?

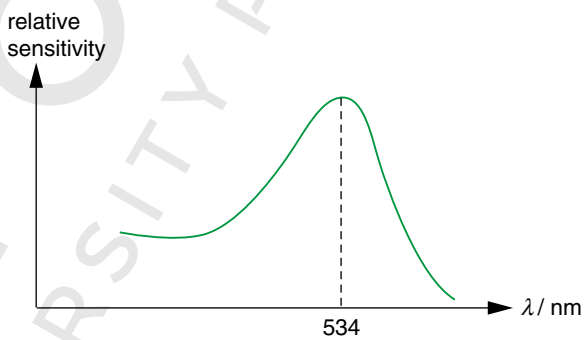


Fig c

★**8** Dorothy is at the bus stop. When a bus approaches, she keeps looking at the route number displayed at the front of the bus (Fig d).



Fig d

(a) Describe the change that takes place in her eye to help her keep seeing the route number clearly.

(b) Using the lens formula, explain why the process in (a) works.

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1.2 Defects of vision and their corrections

Most of us will develop old sight when we grow old.

In Hong Kong, having short-sightedness or long-sightedness is very common. Therefore, it is important to learn about the causes and corrections of these eye defects.



Fig 1.2a *A child wearing a pair of spectacles.*

DSE goal

Define power of a lens and use dioptre as its unit.

DSE exam

19(2)Q4.2

1 Power of a lens

When doing calculations involving defects of vision, it is more convenient to talk about the power of a lens than its focal length.

A convex lens with a shorter focal length f bends light more than one with a longer focal length (Fig 1.2b). We say that the lens with smaller f has greater refractive ability.

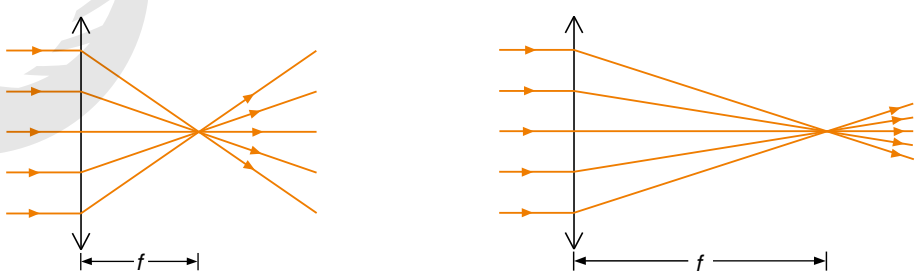


Fig 1.2b *A convex lens with smaller f bends light more while one with larger f bends light less.*

The **power** P of a lens quantitatively measures the refractive ability of a lens. It is defined as

$$P = \frac{1}{f}$$

(P: power of lens (D); f: focal length of lens (m))

1 D = 1 m^{-1} ▶ The unit of the power of a lens is called the **dioptre** (**D**).

We use the + and – signs to distinguish the power for convex and concave lenses respectively (Table 1.2a).

	Convex	Concave
Power	+	–

Table 1.2a *Signs of power.*

✎ Here, we ignore the effect due to the small distance between the lenses.

When a few lenses are placed very close together, their effective power is approximately the sum of the powers of all the lenses. This works for both convex and concave lenses.

The effective power of the lenses placed close together is approximately the sum of the powers of the lenses.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 A lens with a power of (larger / smaller) magnitude refracts light more.

2 Complete the table.

	Lens	Power P	Focal length f	Type of lens
(a)	L_1		0.25 m	
(b)	L_2		–0.8 m	
(c)	L_3	+2 D		
(d)	L_4	–2.5 D		

3 When the following pairs of lenses in Q2 are stacked together, a single lens can be used to replace each pair of lenses and provides the same effect. What is the type and the power of the lens in each case?

(a) L_1 and L_2

(b) L_3 and L_4

(c) L_1 and L_3

power 焦強　　dioptre 屈光度

Learn by practice 2

1 The focal length of a lens is −20 cm.

(a) Find the power of the lens.

(b) Which type of lens does it belong to?

2 The powers of lenses L_1 and L_2 are +3 D and +5 D respectively.

(a) What types of lenses do they belong to?

(b) Which lens refracts light more?

(c) Find the effective power of the lenses when they are placed close together.

3 Two convex lenses of focal lengths 20 cm and 35 cm are placed in contact. Find

(a) the effective power,

(b) the effective focal length

of the combination of lenses.

DSE goal

State the near point and far point of the eye.

DSE exam

14(2)Q.4(b)(i), 20(2)Q4.1

2 Near point and far point

When an object moves towards the eye ($u\downarrow$), the lens needs to become thicker and thicker ($P\uparrow$) to keep the image formed on the retina. The lens will finally reach its maximal thickness when the object gets to a certain position. This position is defined as the **near point** of the eye, which is at a distance of about 25 cm for a normal eye.

What will you see if the object is placed at a position closer to your eye than the near point? ▶

On the other hand, when an object moves away from the eye ($u\uparrow$), the lens needs to become thinner and thinner ($P\downarrow$). The lens reaches its minimal thickness when the object is at the **far point** of the eye. The far point is at infinity (∞) for a normal eye.

Near point

- **the closest point to the eye at which an object can be placed and still form a sharp image on the retina**
- **at about 25 cm for a normal eye**

Far point

- **the furthest point from the eye at which an object can be placed and still form a sharp image on the retina**
- **at infinity for a normal eye**

near point 近點　　far point 遠點

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Locating your near point

What you need



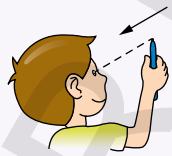
What to do

1



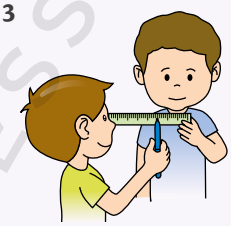
Take off your glasses if you wear them. Close your left eye. Then, hold the pen at a position where you can clearly see the tip of the pen with your right eye.

2



Slowly move the pen closer towards your right eye until the tip of the pen becomes blurred. Move it backwards a little bit to a point where you can see it clearly again.

3



Ask your classmate to measure the distance between the tip of the pen and your right eye.

What you find

How far is the near point from your right eye?

Repeat with your left eye. Is the result the same for both of your eyes?

Example 2 Power of the refracting parts of an eye

For a normal eye, the range of accommodation is from 25 cm to infinity. The smallest **power of the refracting parts** of the eye is +40 D.

(a) What happens to the power of the lens when the eye is looking at an object at its near point and far point respectively?

(b) Find the distance d between the retina and the optical centre of the refracting parts of the eye.

Solution

(a) When the eye is looking at an object at its near point, the power of the lens is the largest.

When the eye is looking at an object at its far point, the power of the lens is the smallest.

(b) Apply $P = \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$. When $u = \infty$, $P = +40$ D,

$$40 = \frac{1}{\infty} + \frac{1}{d}$$

$$d = 0.025 \text{ m } (2.5 \text{ cm})$$

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 An eye cannot see an object closer than the near point **clearly** because
- the lens of the eye cannot become (thin / thick) enough, or
- the power of the lens cannot become (small / large) enough,
- the light from the object is converged too (little / much).

2 An eye cannot see an object further than the far point **clearly** because
- the lens of the eye cannot become (thin / thick) enough, or
- the power of the lens cannot become (small / large) enough.
- the light from the object is converged too (little / much).

3 The far point of a normal eye is at _____.

4 An object is placed at a distance d in front of a lens with power P. Find the distance x of the image from the lens and determine whether the image is real or virtual.

	d	P	x	Real or virtual
(a)	0.5 m	+3 D		
(b)	8 cm	+20 D		
(c)	20 cm	−10 D		

Learn by practice 3

1 The power of the refracting parts of an eye is +42 D when looking at an object 50 cm in front of it. What is the distance between the retina and the optical centre of the refracting parts?

2 The distance between the retina and the optical centre of the refracting parts of an eye is 2.5 cm.

(a) If the eye is looking at an object at 2 m away, what is the power of the refracting parts of the eye?

(b) The power of the refracting parts is +41.6 D when the eye is looking at an object at a distance d. Find d.

3 The distance between the retina and the optical centre of the refracting parts of an eye is 2.8 cm. The near point and the far point of the eye are at 40 cm and at infinity respectively. Find the minimum power and maximum power of the refracting parts of the eye.

4 The distance between the retina and the optical centre of the refracting parts of an eye is 2.4 cm. The minimum power and maximum power of the eye are +42 D and +46 D respectively. Find the near-point distance and the far-point distance of the eye.

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DSE goal

Describe some defects of vision and their corrections.

DSE exam

14(2)Q4.1, 15(2)Q4.1, 16(2)Q4.2, 17(2)Q4.1, 18(2)Q4.2, 19(2)Q4.1, 22(2)Q4.1, 23(2)Q.4(b)&(c)

→ This video demonstrates Expt 1b, which is time-consuming and difficult to set up.

Experiment 1b

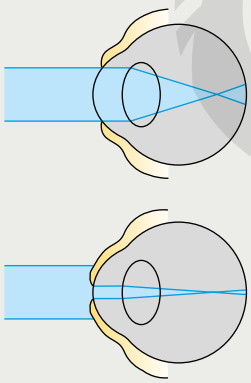
The model eye and defects of vision (p.184) Video

Starting from here, most of the ray diagrams would be simplified. Only the refraction of light rays at the front surface of the lens would be shown. The refraction at the surface of the cornea and that at the back surface of the lens would be omitted.

Know more

Half-closed eye

'Myopia' means 'half-closed eye' in Greek. If you have short-sightedness, you can see distant objects more clearly by half-closing your eye. This is because the light beam entering your eye becomes narrower and hence less spread out on the retina.



3 Defects of vision and their corrections

The far point and the near point of a normal eye are respectively at infinity and at 25 cm. However, some people's eyes have closer far points or further near points. They may not be able to see far objects or near objects clearly. Such people have defects of vision.

a Short-sightedness (myopia)

i What is short-sightedness?

On the other hand, people suffering from short-sightedness can see near objects clearly.

People suffering from **short-sightedness** (also called **myopia**) cannot see far objects clearly. This is either because the eyeball is too long, or the lens is too thick when the ciliary muscles relax.

For these people, light from a far object is converged too much, even when the lens is at its minimal thickness (Fig 1.2c), i.e. at its smallest power. The image is formed in front of the retina, making the object look blurred. The far point of the eye is closer to that of a normal eye (at infinity).

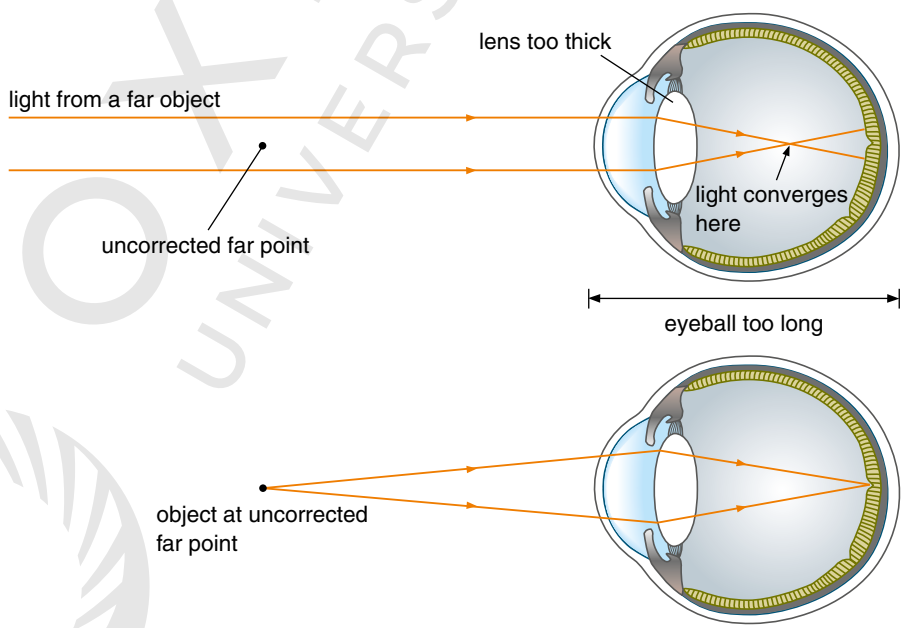


Fig 1.2c Short-sightedness.

ii Correction

The use of diverging lens can be explained in another way: The diverging lens is used to reduce the converging effect of the over-powered lens of the eye.

Short-sightedness can be corrected by using spectacles with diverging (concave) lenses. In order to correct the far point to infinity, **the diverging lens should be able to form a virtual image of a distant object at the uncorrected far point of the eye** (Fig 1.2d on p.27). This virtual image acts as an object for the eye. As a result, an image can be formed on the retina.

short-sightedness / myopia 近視

The figure shows the case in which the object is at infinity. For an object closer than infinity, the virtual image is formed closer than the uncorrected far point of the eye.

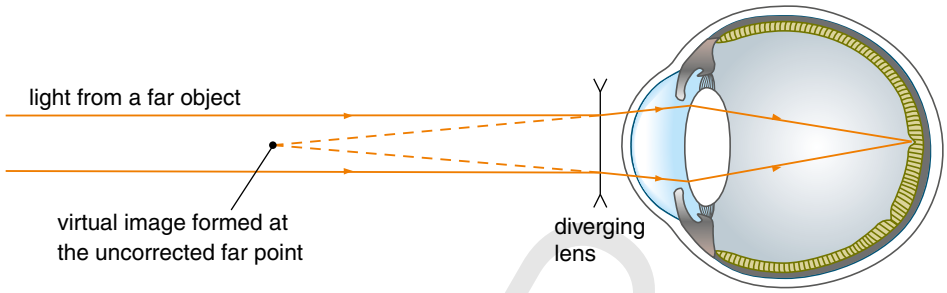


Fig 1.2d *Correcting short-sightedness with a diverging lens.*

b Long-sightedness (hypermetropia)

i What is long-sightedness?

On the other hand, people suffering from long-sightedness can see far objects clearly.

People suffering from **long-sightedness** (also called **hypermetropia**) cannot see near objects clearly. This is either because the eyeball is too short, or the lens is too thin when the ciliary muscles contract maximally.

In both situations, the power of the lens is too low for the eye even at its maximal thickness. Light from a near object is converged too little and converges to somewhere behind the retina, making the object look blurred (Fig 1.2e). In other words, the near point of a long-sighted eye is further away than the typical value of 25 cm.

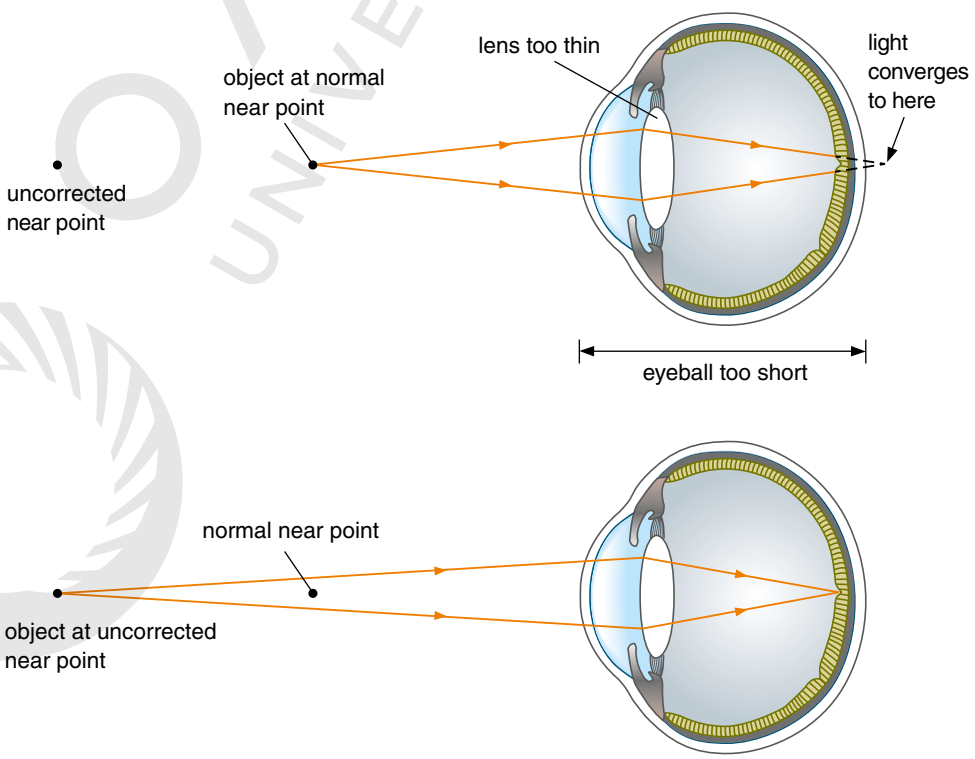


Fig 1.2e *Long-sightedness.*

long-sightedness / hypermetropia 遠視

ii Correction

✎ The use of converging lens can be explained in another way:
The converging lens is used to increase the converging effect of the under-powered lens of the eye.

Long-sightedness can be corrected by using spectacles with converging (convex) lenses. In order to correct the near point to 25 cm, **the converging lens forms a virtual image of a near object located at 25 cm at the uncorrected near point of the eye** (Fig 1.2f). This virtual image acts an object for the eye. As a result, an image can be formed on the retina.

The figure shows the case in which the object is at the normal near point. For an object further away from the normal near point, the virtual image is formed further away than the uncorrected near point of the eye.

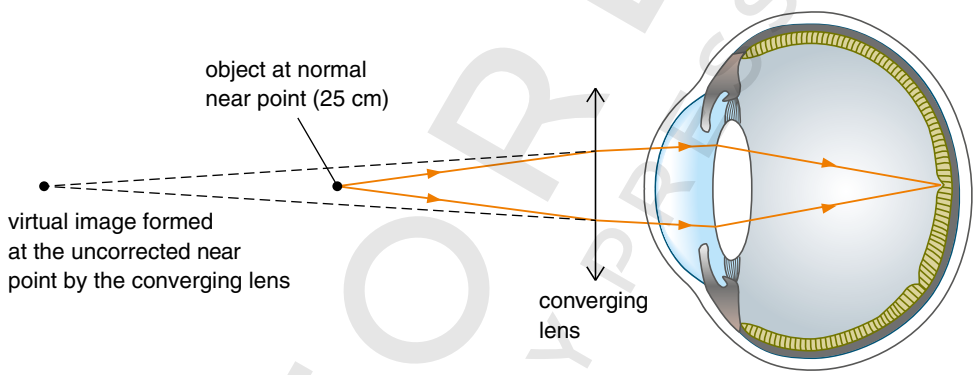


Fig 1.2f *Correcting long-sightedness with a converging lens.*

C Old-sightedness (presbyopia)

i What is old-sightedness?

✎ Old-sightedness can be regarded as a kind of long-sightedness related to ageing.
✎ It appears that for many people there is a contest between short-sightedness and old-sightedness throughout life.

During accommodation, the ciliary muscles contract to allow the lens to thicken under its own elasticity. As we get older, the lens will thicken less because of its decreased elasticity. This reduces the eye's power and hence its ability for accommodation.

In fact, the decrease in accommodation ability begins at about 10 years old. It is usually noticed between the ages of 40–50 and stabilizes at the age of about 60.

An older person's eyes are only able to accommodate to a limited extent. When seeing a close object, e.g. when reading, light from the object will not converge on the retina but behind it instead (a situation similar to long sight) (Fig 1.2g). This is called **old-sightedness** (also called **presbyopia**).

✎ As old-sightedness deepens, the near-point distance increases. The person has to hold the object (book, newspaper) further and further away to compensate.

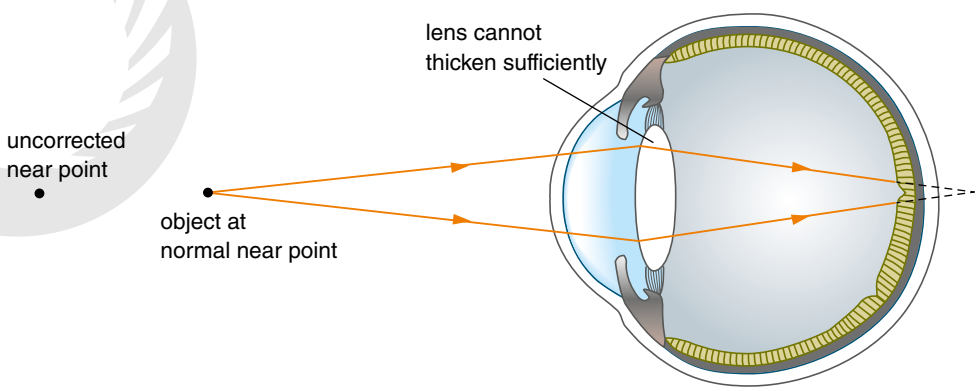


Fig 1.2g *Old-sightedness.*

old-sightedness / presbyopia 老花

ii Correction

Similar to the case of long-sightedness, spectacles with converging lenses can be used to help an old-sighted eye see near objects clearly (Fig 1.2h).

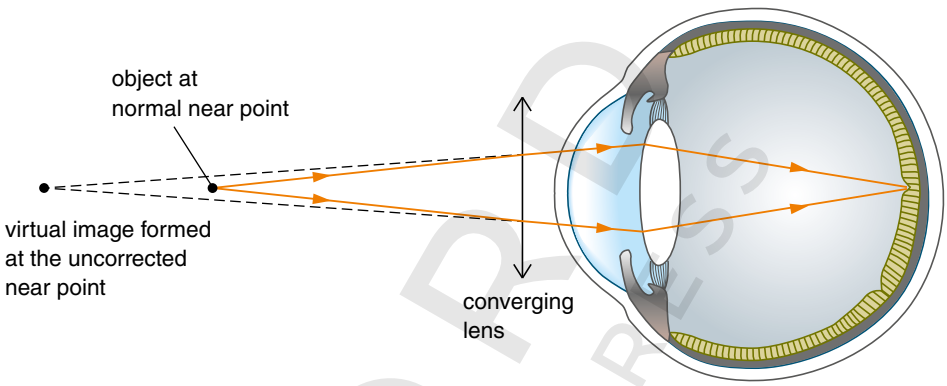


Fig 1.2h *Correcting old-sightedness with a converging lens.*

	Short-sightedness	Long-sightedness	Old-sightedness
Problem	cannot see far objects clearly	cannot see near objects clearly	cannot see near objects clearly
Cause	eyeball too long or lens too thick	eyeball too short or lens too thin	elasticity of lens decreases due to ageing
Correction	diverging lens	converging lens	converging lens

Note that some people suffering from old-sightedness also suffer from short- or long-sightedness. People suffering from both long- and old-sightedness can wear spectacles with converging lenses for correction, but people suffering from both short- and old-sightedness may need to wear bifocals, with each lens consisting of parts of two distinct powers (Fig 1.2i).



Fig 1.2i *Bifocals.*

Google Forms Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* A person suffering from long-sightedness can see an object located between the near point and the far point of his eye clearly. (T/F)

2 Complete the table below to show how the near-point distance and far-point distance of eyes with defects are different from those of a normal eye.

	Near-point distance	Far-point distance
Normal eye	d, typically 25 cm	∞
Short-sighted eye	(d / > d / < d)	(∞ / < ∞)
Long-sighted eye	(d / > d / < d)	(∞ / < ∞)

3 For a **short-sighted eye**,

- light is converged too (little / much), or
- the power of the refracting parts is too (small / large)

when the eye is seeing (near / far) objects.

4 Spectacles with (converging / diverging) lenses help correct a **short-sighted eye** by

- (converging / diverging) the light, or
- (increasing / decreasing) the effective power.

5 For a **long-sighted eye**,

- light is converged too (little / much), or
- the power of the refracting parts is too (small / large)

when the eye is seeing (near / far) objects.

6 Spectacles with (converging / diverging) lenses help to correct a **long-sighted eye** by

- (converging / diverging) the light, or
- (increasing / decreasing) the effective power.

7 Johnny suffers from **short-sightedness** and wears a pair of spectacles to correct this defect. Let P be the uncorrected far point. Which of the following is / are correct?

☐ A distant object forms an image at P by the spectacles.

☐ An object at P forms an image at infinity by the spectacles.

8 Jenny suffers from **long-sightedness** and wears a pair of spectacles to correct this defect. Let P be the uncorrected near point and Q be the corrected near point. Which of the following is / are correct?

☐ An object at P forms an image at Q by the spectacles.

☐ An object at Q forms an image at P by the spectacles.

9 An old-sighted eye is unable to see a (near / far) object clearly, similar to the situation of a (short-sighted / long-sighted) eye.

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Learn by practice 4

1 (a) How are the near point and far point of a short-sighted eye different from those of a normal eye?

(b) Suggest two possible causes for the difference in (a).

2 (a) How are the near point and far point of a long-sighted eye different from those of a normal eye?

(b) Suggest two possible causes for the difference in (a).

3 (a) How are the near point and far point of an old-sighted eye different from those of a normal eye?

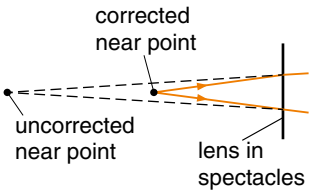
(b) Suggest a possible cause for the difference in (a).

When we solve problems on correcting eye defects with spectacles, we usually need to apply the lens formula:

$$P = \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

We can apply it by considering one of the following three situations:

An example:



▶ **1** An object at the **corrected near/far point** forms an image at the **uncorrected near/far point** by the **spectacles**.

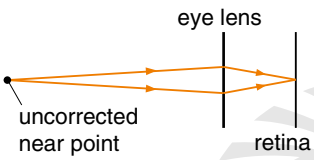
power of spectacles (D)

$$P = \frac{1}{u} + \frac{1}{v}$$

corrected near/far-point **distance** (m)

uncorrected near/far-point **distance** (m)

An example:



▶ **2** An object at the **uncorrected near/far point** forms an image on the **retina** by the **uncorrected eye**.

power of eye lens (D)

$$P = \frac{1}{u} + \frac{1}{v}$$

uncorrected near/far-point **distance** (m)

distance between retina and optical centre of refracting parts of eye (m)

The corrected eye means the whole combination of the spectacles and the eye.

▶ **3** An object at the **corrected near/far point** forms an image on the **retina** by the **corrected eye**.

effective power of corrected eye (D)

$$P = \frac{1}{u} + \frac{1}{v}$$

corrected near/far-point **distance** (m)

distance between retina and optical centre of refracting parts of eye (m)

Example 3 | Correction for short sight

Jason is suffering from short-sightedness and is wearing spectacles with lenses of power −1.25 D to correct the defect.

(a) Figures a and b show the case without and with the spectacles. Complete the paths of the light rays to show how they reach the retina.

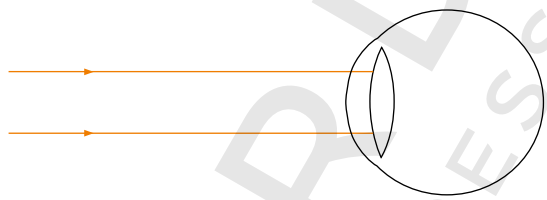


Fig a

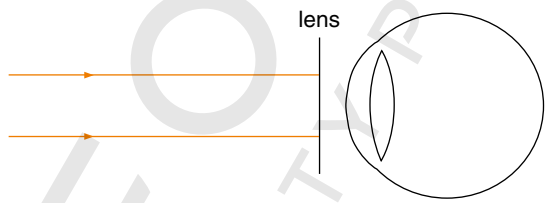
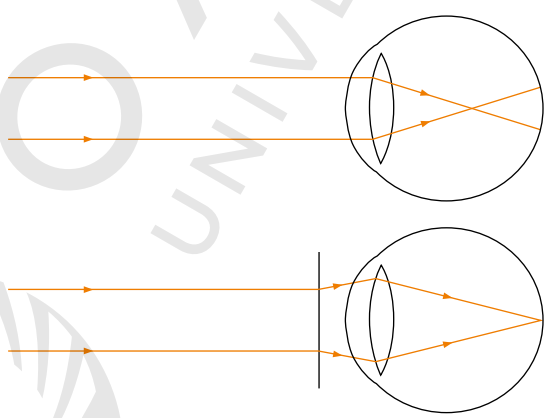


Fig b

(b) Find the distance of his uncorrected far point from his eyes.

Solution

(a)



(b) A distant object ($u = \infty$) forms an image at the uncorrected far point (v = ?) by the spectacles (P = −1.25 D).

$$P = \frac{1}{u} + \frac{1}{v}$$

$$-1.25 = \frac{1}{\infty} + \frac{1}{v}$$

$$v = -0.8 \text{ m}$$

His uncorrected far point is 0.8 m from his eye.

Example 4 Correction for long sight

The power P_0 of a long-sighted eye is +50.25 D when it is looking at an object at its near point. The distance between the retina and the optical centre of the refracting parts of the eye is 2 cm.

(a) After wearing spectacles, the near point is corrected to 25 cm. Find the power P_1 of the lenses of the spectacles.

(b) Find the effective power P_{eff} of the corrected eye when looking at an object at its near point.

Solution

(a) An object at the uncorrected near point (u = ?) forms an image on the retina (v = 0.02 m) by the uncorrected eye (P = +50.25 D).

$$P = \frac{1}{u} + \frac{1}{v}$$

$$+50.25 = \frac{1}{u} + \frac{1}{0.02}$$

$$u = 4 \text{ m}$$

The uncorrected near point is 4 m from the eye.

An object at the corrected near point (u = 0.25 m) forms an image at the uncorrected near point (v = −4 m) by the spectacles ($P = P_1$).

$$P = \frac{1}{u} + \frac{1}{v}$$

$$P_1 = \frac{1}{0.25} + \frac{1}{-4}$$

$$P_1 = +3.75 \text{ D}$$

(b) $P_{eff} = P_0 + P_1$

$$= 50.25 + 3.75$$

$$= +54 \text{ D}$$

Think about...

Can you find P_{eff} without finding P_1?

Know more

'Degree' of short sight and long sight

In Hong Kong, we are often diagnosed to have a certain 'degree' of short/long sight. 'Degree' is a smaller unit of refractive power (1 D = 100 degrees). A short-sighted eye which is corrected by a lens of −1.25 D has a short sight of 125 degrees.

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Quick check 8

→ E-version in OUP Exercise Platform is also available.

1 Complete the table for **short-sighted eyes**.

	Uncorrected far-point distance	Corrected far-point distance	Power of corrective spectacles
(a)	1.2 m	∞	
(b)		∞	−1.5 D

2 Complete the table for **long-sighted eyes**.

	Uncorrected near-point distance	Corrected near-point distance	Power of corrective spectacles
(a)	80 cm	25 cm	
(b)		25 cm	+1.5 D

3 Complete the table for **short-sighted eyes**.

	Uncorrected far-point distance	Power of eye when looking at object at uncorrected far point	Distance between retina and optical centre of refracting parts of eye
(a)	1.2 m		2 cm
(b)		+50.1 D	2.05 cm
(c)	80 cm	+51.5 D	

4 Complete the table for **long-sighted eyes**.

	Uncorrected near-point distance	Power of eye when looking at object at uncorrected near point	Distance between retina and optical centre of refracting parts of eye
(a)	45 cm		2 cm
(b)		+52.5 D	2.02 cm
(c)	60 cm	+51.5 D	

5 Complete the table for **short-sighted eyes**.

	Minimum power of eye	Minimum power of corrected eye	Required power for corrective spectacles
(a)	+49.9 D	+48.8 D	
(b)		+49.2 D	−2.5 D

6 Complete the table for **long-sighted eyes**.

	Maximum power of eye	Maximum power of corrected eye	Required power for corrective spectacles
(a)	+50.8 D	+53.2 D	
(b)		+54.1 D	+2 D

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Learn by practice 5

1 The far point of a short-sighted eye is at 2.5 m. Find the power of the spectacles required to correct the defect.

2 A pair of spectacles is used to correct the near point of an eye from 40 cm to 25 cm. What is the power of the spectacles?

3 Samson is suffering from long-sightedness. He needs a pair of spectacles with a power of +1.2 D to correct his near-point distance to 25 cm. Find his near-point distance before wearing the spectacles.

4 Joyce sees an object 50 cm in front of her eye clearly and the power of her eye is +48 D at that time. Find the distance between the retina and the optical centre of the refracting parts of her eye.

5 It is given that the near point of an eye is at 25 cm and the maximum power of the eye is +52 D.

(a) Find the distance between the retina and the optical centre of the refracting parts of the eye.

(b) The far point of the eye is at infinity. Find the minimum power of the eye.

6 The near point of an old-sighted eye is at 50 cm and the distance between its retina and the optical centre of its refracting parts is 2 cm.

(a) What is the power of the eye when it is looking at an object at the near point?

(b) Find the power of the spectacles required to correct the near point to 25 cm.

7 Joe is suffering from short-sightedness and is wearing spectacles with a power of –2.5 D to correct the defect. The distance between the retina and the optical centre of the refracting parts is 2.18 cm.

(a) Find the uncorrected far-point distance.

(b) Find the effective power of Joe's eye with the spectacles when looking at a distant object.

Know more

Astigmatism

Astigmatism occurs when the cornea is irregular in shape. A person with astigmatism sees some parts of an object more clearly than other parts. For example, the person sees the lines in one direction in the figure more sharply in focus than the lines in other directions. It is corrected by lenses with a curvature that compensates for the irregular shape of the cornea.

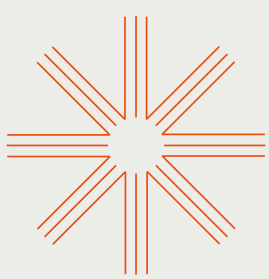


Chart for checking astigmatism.

astigmatism 散光

☆ HKDSE 2019 Paper 2 Q4.1

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Effect on near point and far point

Jason is suffering from short-sightedness. His near point and far point are at 20 cm and 2 m respectively. What will be his near point if he wears spectacles to correct the short-sightedness?

A 18.2 cm

B 20 cm

C 22.2 cm

D 25 cm

Solution

The spectacles will correct the far point to infinity.

Suffering from short-sightedness means there is a problem with the far point. The power of the spectacles should be calculated by considering the correction of the far point (but not the near point).

A distant object ($u = \infty$) forms an image at the uncorrected far point ($v = -2$ m) by the spectacles (P = ?).

$$P = \frac{1}{u} + \frac{1}{v} = \frac{1}{\infty} + \frac{1}{-2} = -0.5 \text{ D}$$

An object located at the uncorrected near point ($u = 0.2$ m) forms an image at the corrected near point (v = ?) by the spectacles ($P = -0.5$ D).

$$P = \frac{1}{u} + \frac{1}{v}$$

$$-0.5 = \frac{1}{0.2} + \frac{1}{v}$$

$$v = -0.182 \text{ m}$$

His near point will be at 0.182 m.

Common mistake ⚠
Students may wrongly think that spectacles for correcting short-sightedness will not cause any change to the near point.

∴ The answer is A.

▶ Revision 1 Q15 (p.53)

Exercise 1.2

1 Which of the following statements about short-sightedness is **incorrect**?

A A short-sighted person cannot see far objects clearly.

B The image of a far object is formed in front of the retina of a short-sighted eye.

C The far point of a short-sighted eye is not at infinity.

D Short-sightedness can be corrected by wearing spectacles with converging lenses.

***2**

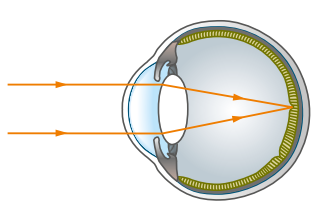


Fig a

Which of the following is a possible description of the above ray diagram?

A A short-sighted eye looking at a near object

B A short-sighted eye looking at a distant object

C A long-sighted eye looking at a near object

D A long-sighted eye looking at a distant object

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*3 Which of the following is/are the cause(s) of old-sightedness?

(1) The elasticity of the lens decreases with age.

(2) The length of the eyeball decreases with age.

(3) The lens is too thick even though the ciliary muscles are relaxed.

A (1) only B (1) and (2) only

C (2) and (3) only D (1), (2) and (3)

4 Describe what the near point and the far point of an eye are. State their positions for a normal eye.

5 The near point of an eye is at 25 cm and its minimum focal length is 2.5 cm.

(a) Find the distance between the retina and the optical centre of the refracting parts of the eye.

(b) The far point of the eye is at infinity. Find the minimum power of the eye.

*6 When Ivy looks at an object 40 cm in front of her, the lenses of her eyes are at their maximal thickness. At that moment, the power of her eye is +42.5 D.

(a) Can she read the text on a book clearly if the book is at 30 cm in front of her? Explain briefly.

(b) Find the distance between the retina and the optical centre of the refracting parts of her eye.

*7 The figures below show a long-sighted eye without and with a suitable corrective lens respectively. Complete the paths of the light rays to show how they reach the retina.

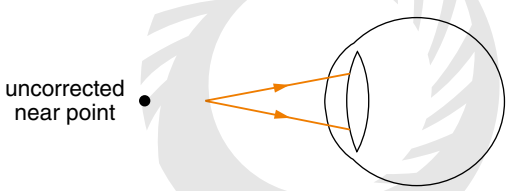


Fig b

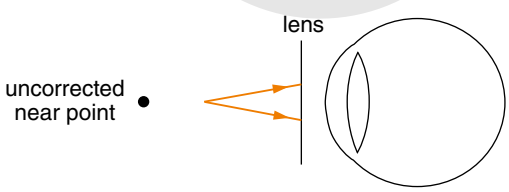


Fig c

*8 A diverging lens has a focal length of 0.4 m.

(a) Find the power of the lens.

(b) The lens is now placed in contact with another lens of power –2 D.

(i) What is the effective power of the combined lens system?

(ii) Is the combined lens system convergent or divergent?

(iii) An object is placed 0.5 m away from the optical centre of the combined lens system. What will be the position and the nature of the image formed?

*9 Jamie is suffering from short-sightedness. He needs a pair of spectacles with a power of –2.5 D to correct the defect. His near-point distance is 25 cm.

(a) Find his uncorrected far-point distance.

(b) Find his near-point distance after wearing the spectacles.

*10 Peggy is suffering from a defect of vision and is wearing a corrective lens with a power of +1.5 D. The near point of her eye without the corrective lens is at 40 cm.

(a) (i) What type of lens is the corrective lens?

(ii) Suggest a possible eye defect that Peggy is suffering from.

(b) Draw a ray diagram to show how Peggy's eye defect is corrected.

(c) Suppose an object is placed 30 cm in front of the corrective lens. Find the position of the image formed by the corrective lens.

*11 Peter is suffering from short-sightedness. He wears a pair of spectacles with a power of –2 D to correct the defect. When he is wearing the spectacles, the near point of his eye is at 30 cm. Find his near-point distance and far-point distance without wearing the spectacles.

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1.3 Fibre optic endoscopy

In this part, we will look at a medical imaging device called **fibre optic endoscope** (Fig 1.3a), which can be used to look inside the various cavities of the body (Fig 1.3b). The use of a fibre optic endoscope is called **fibre optic endoscopy**.



Fig 1.3a *A fibre optic endoscope.*



Fig 1.3b *Looking inside a patient's body with an endoscope.*

DSE goal

Describe how a fibre optic endoscope works.

1 Introduction to fibre optic endoscopy

A fibre optic endoscope has a thin flexible tube that can be inserted into a patient's body during the imaging process (Fig 1.3c). It consists of optical fibres which allow light to travel from one end to the other by total internal reflection. The image of the body part formed at one end can then be reproduced and observed by doctor at the other end.

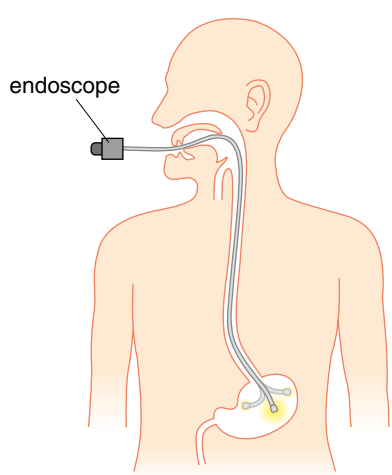


Fig 1.3c *Observing the inner surface of stomach with an endoscope.*

fibre optic endoscope 光纖內窺鏡　　fibre optic endoscopy 光纖內窺鏡檢查術

2 Bundles of optical fibre

The special design of each optical fibre and the special arrangement of the optical fibres in a bundle make it possible for the fibre optic endoscope to transmit images from cavities inside the body.

a Characteristics of an optical fibre

DSE goal

Describe the characteristics of an optical fibre.

Solve problems involving optical fibre.

DSE exam

18(2)Q4.5, 19(2)Q.4(a)(i)&(ii), 23(2)Q4.1

An optical fibre is made of a glass (or polymer) core surrounded by a **cladding** with a smaller refractive index. This allows light to travel along the core by a series of total internal reflections (Fig 1.3d). In addition, the cladding protects the core against scratches and dirt.

✎ The cladding also prevents adjacent fibres from touching each other. Light may escape from one fibre into another at the points of contact of the fibres.

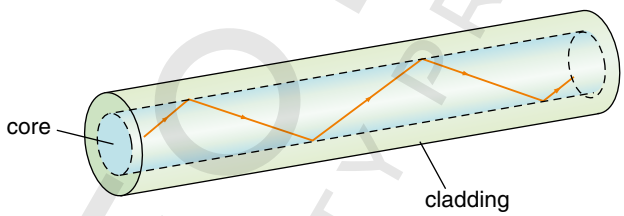


Fig 1.3d *A series of total internal reflections of light at the interface between the core and the cladding.*

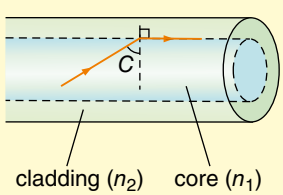
Recall from Book 3A Chapter 2 that total internal reflection occurs when a light ray travels from a medium with larger refractive index n_1 to a medium with smaller refractive index n_2 at an angle of incidence greater than the critical angle C. The critical angle can be found by Snell's law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

When $\theta_1 = C$, $\theta_2 = 90°$,

$$n_1 \sin C = n_2 \sin 90°$$

refractive index of core — critical angle (°) — refractive index of cladding



cladding 包覆層

An optical fibre can function normally when bent slightly (Fig 1.3e), but when the degree of bending is too large, light may escape (Fig 1.3f). A thinner fibre allows a larger degree of bending without the escape of light (Fig 1.3g). Therefore, a fibre is usually made ultra-thin.

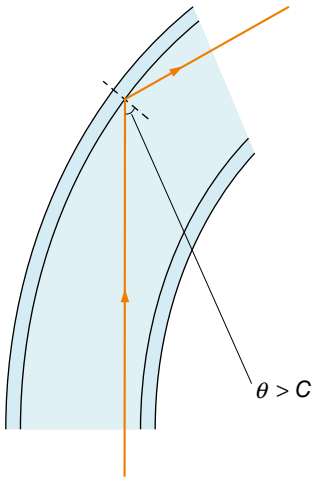


Fig 1.3e *Light travelling in a slightly bent fibre.*

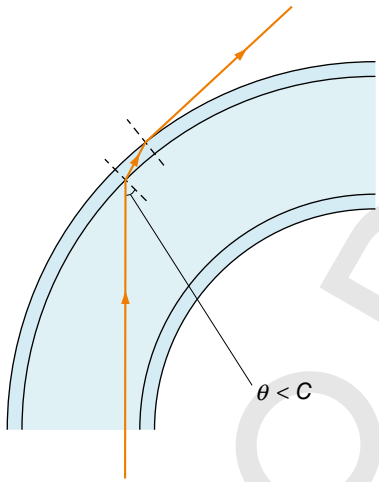


Fig 1.3f *Light escapes from a bent fibre.*

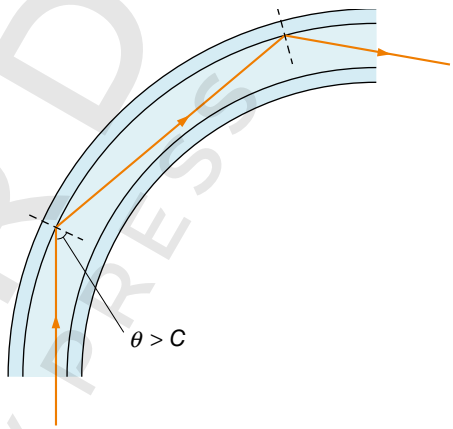
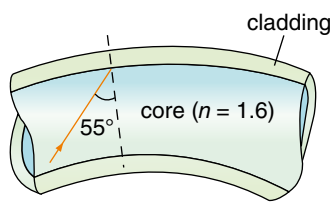


Fig 1.3g *Light is less likely to escape in a thinner fibre.*

There are 3 concepts here:
1 Ultra-thin fibres enhance total internal reflection
2 Cladding enhances total internal reflection
3 Cladding protects the core

Example 5 Refractive index of cladding in an optical fibre

The figure shows an optical fibre consisting of an inner core with refractive index 1.6 and an outer cladding. The minimum angle of incidence for a light ray not escaping at the interface between the core and the cladding is 55°.



Find the refractive index of the cladding.

Solution

By Snell's law,

$n_1 \sin \theta_1 = n_2 \sin \theta_2$

$1.6 \sin 55° = n_2 \sin 90°$

$n_2 = 1.31$

The refractive index of the cladding is 1.31.

Think about...

Part of the optical fibre is immersed in oil which has a refractive index of 1.47. Will this affect the total internal reflection of light rays in the optical fibre? What if the optical fibre has no outer cladding?

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 An optical fibre is made of a core with a (smaller / larger) refractive index surrounded by a cladding with a (smaller / larger) refractive index. Light travels within the core by a series of (refraction / diffraction / total internal reflection).

2 Complete the table below.

	n_1 (X)	n_2 (Y)	Can total internal reflection occur when light travels from media X to Y?	Critical angle (if total internal reflection can occur)
(a)	1.52	1.15		
(b)	1.24	1.52		
(c)	1.52		Yes	72.5°
(d)		1.28	Yes	45.6°

3 Bending of optical fibre may cause the escape of light. A (thinner / thicker) optical fibre allows a larger degree of bending.

Learn by practice 6

1 The core of an optical fibre has a refractive index of 1.75 and the outside cladding has a refractive index of 1.55 (Fig a). Find the minimum value of p for the light ray in the optical fibre to be totally internally reflected.

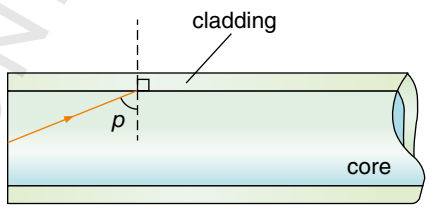


Fig a

2 The core of an optical fibre has a refractive index of 1.66. A light ray enters the core and travels in it as shown (Fig b). Find the values of q and r.

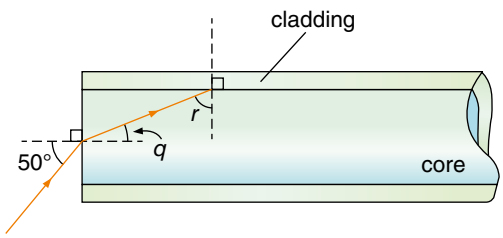


Fig b

DSE goal

Describe how coherent bundle fibres form image.

DSE exam

12(2)Q4.2, 13(2)Q4.3

b Coherent and incoherent bundle fibres

In an endoscope, light is transmitted by bundles of optical fibres. There are two types of bundles in an endoscope, one for image transmission and one for illumination.

i Coherent bundle fibres

A **coherent bundle fibre** is manufactured in a way that the relative positions of the fibres are preserved from one end to the other (Fig 1.3h). This maintenance of position allows an image at one end to be reproduced faithfully at the other end.

By using thinner optical fibres to make the bundle fibre, an image with higher resolution can be produced (Fig 1.3i).

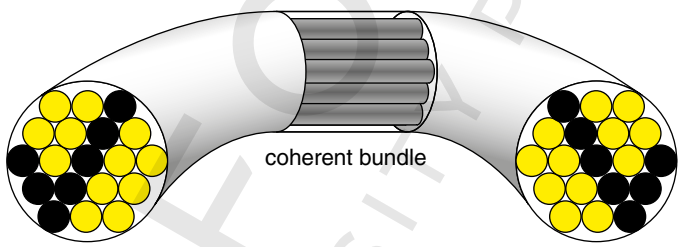


Fig 1.3h *The pattern of the dots making up the image is preserved.*

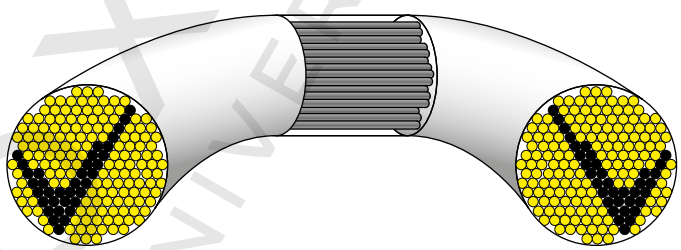


Fig 1.3i *The image has a higher resolution if a bundle with thinner optical fibres is used.*

Know more

How does the brain recognize an image?

The brain recognizes an image by identifying the patterns in the colour dots picked up by the retina (Fig a). Therefore, the key for transmitting an image is to preserve the patterns of the dots that form the image.

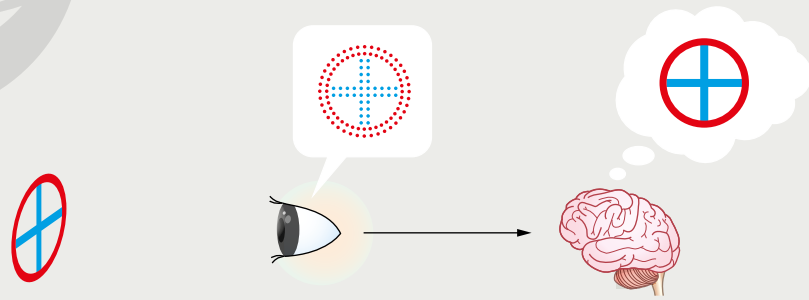


Fig a

coherent bundle fibre 相干光纖管束

ii Incoherent bundle fibres

An **incoherent bundle fibre** is manufactured in a way that the relative positions of the fibres do not need to be preserved (Fig 1.3j) and therefore cannot be used to transmit images. It is used to transmit light into the body for illumination. They are cheaper and easier to manufacture.

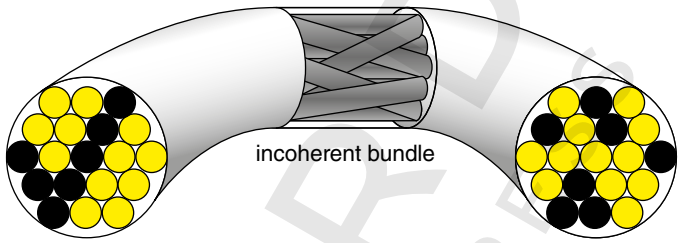


Fig 1.3j *An incoherent bundle fibre cannot be used to transmit images.*

- **A coherent bundle fibre preserves the relative position of each fibre from one end to the other. It is used to transmit images.**
- **An incoherent bundle fibre is used to transmit light for illumination.**

C Role of bundle fibres in an endoscope

When using an endoscope, the flexible tube and the bending section (Fig 1.3k) are inserted into the patient's body through an opening in the body, i.e. mouth, nose, anus, or a small hole cut in the skin. Light is transmitted by a light guide (incoherent bundle) into the body. The light is then reflected by the interior surface of the cavity being examined and collected by the objective lens to form an image on one end of the coherent bundle. The image is transmitted by the coherent bundle to the other end, and is magnified by the eyepiece for the doctor to see.

With additional connections, the image can also be displayed on a computer screen.

Note that **light** can travel from the objective lens to the eyepiece or from the eyepiece to the objective lens. ▶

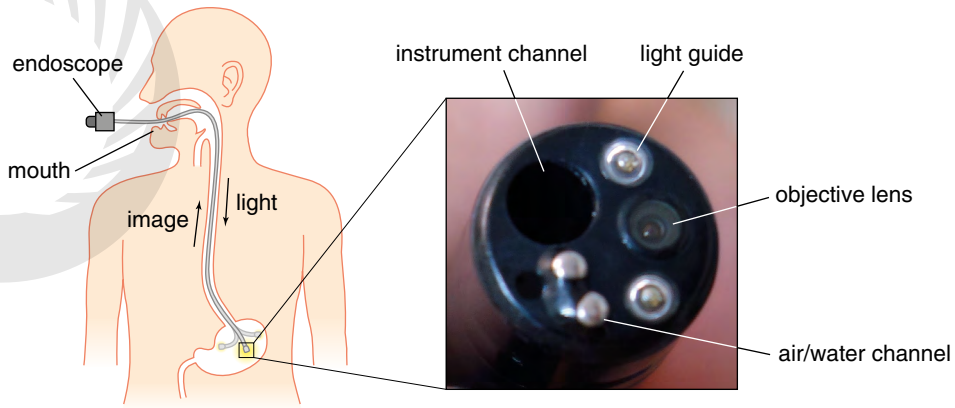


Fig 1.3k *At the tip of the bending section, there is an objective lens (at the entry of the coherent bundle), an entry of the light guide and some other channels.*

Know more

Structure of a fibre optic endoscope

Do you want to know more about the structure of a fibre optic endoscope?

▶ p.58

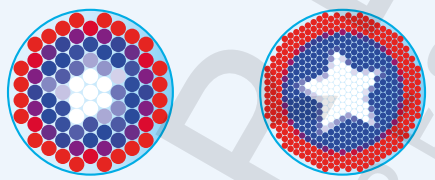
incoherent bundle fibre 不相干光纖管束

Right and wrong

Brightness of image

✗ For two coherent bundle fibres with the same diameter, the one made with thinner optical fibres produces a much brighter image than the one with thicker optical fibres.

✓ The total amount of light transmitted by the two coherent bundle fibres is about the same. Therefore, they produce an image of about the same brightness.



bundle with thicker optical fibres

bundle with thinner optical fibres

Google Forms

Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 An image can be transmitted through (coherent / incoherent) bundle fibres where the relative positions of the fibres are (preserved / not preserved) from one end to another.

2 A coherent bundle with thinner fibres produces image with (lower / higher / about the same) resolution and (lower / higher / about the same) brightness.

3 During the operation of an endoscope, (light / image) is transmitted into the body by (coherent / incoherent) bundle fibres and the (light / image) produced is transmitted to the outside by (coherent / incoherent) bundle fibres.

4 Arrange the following processes in order.

a Image is transmitted by a coherent bundle.

b Light is transmitted by the light guide (incoherent bundle) into the body.

c Light is reflected by the body part.

d The doctor sees the image through the eyepiece.

e Image is formed by the objective lens.

DSE goal

Discuss the advantages and limitations of endoscopy.

DSE exam

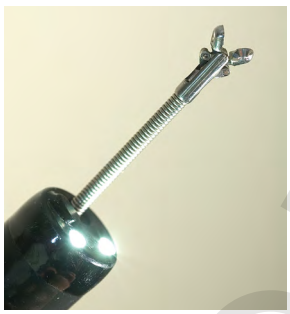
14(2)Q4.4, 15(2)Q4.2, 17(2)Q4.6, 22(2)Q4.2

3 Usages, advantages and limitations of fibre optic endoscopy

a Uses of fibre optic endoscopy

Since the bending section of an endoscope is flexible, it allows doctors to look around corners inside the patient's body to facilitate diagnosis. Therefore, an endoscope is usually used to inspect the inner surface of any hollow organ, such as the digestive tract, the urinary bladder and the uterus.

In addition to imaging, the endoscope allows the doctor to pass other instruments through the endoscope shaft. For example, a pair of tiny forceps can be passed to the tip to obtain a tissue sample from the inner parts of the body for further examination (Fig 1.3l). This allows doctors to confirm directly the suspected occurrence of tumors, inflammation and other abnormalities.



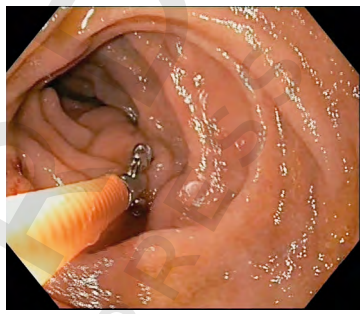


Fig 1.3l *A pair of tiny forceps extended from the tip of the endoscope to draw a tissue sample from the body.*

As the incisions into the body during minimally invasive surgery are small, the time taken for the surgical wound to heal is shorter and the risk of infection is lower than in open surgery.

▶ Endoscopes are also used to perform *minimally invasive surgery*, in which the endoscope enters the body via an artificially created *orifice* of limited size to perform surgery (Fig 1.3m).

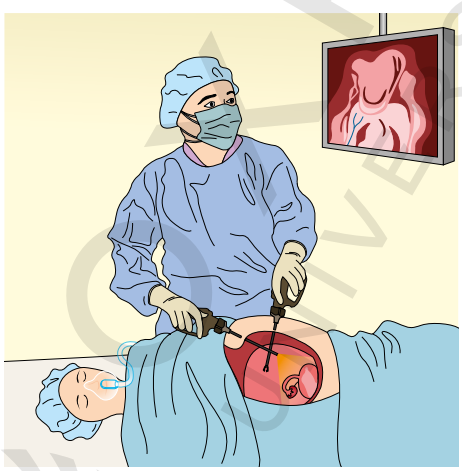




Fig 1.3m *Minimally invasive surgery.*

b Advantages and limitations

Fibre optic endoscopy has the following advantages and limitations.

i Advantages

1 **Direct observation of the organ's surface** can be made. The image observed is of the same colour as the organ's surface.

Imaging methods which involve ionizing radiation will be discussed in Chapter 3.

▶ 2 It **does not involve ionizing radiation**.

3 **Suspicious tissue can be sampled** (taken out of the body) at the same time for further examination.

minimally invasive surgery 微創手術　　orifice 孔

ii Limitations

1 The **field of view is limited** (Fig 1.3n).

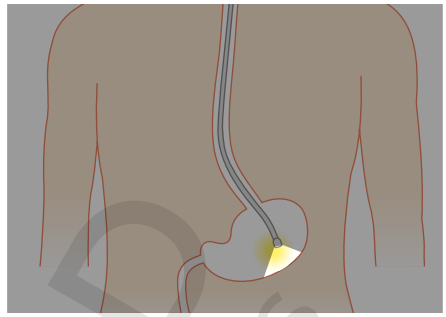


Fig 1.3n *An endoscope can only view a small part of the body at a time.*

This makes the patient less uncomfortable during the process.

2 ***Anaesthesia* is often required** before the examination.

3 Fasting is needed before examining the digestive tract to empty the bowels.

We are only able to look at the surfaces of an organ but not the internal structures.

4 **Only the inner surfaces of hollow organs can be examined**, e.g. stomach and colon.

5 **Perforation of the organ may occur** due to over-inflating the organ with air or deep tissue sampling.

6 **Complications may arise** if the endoscope is contaminated.

7 **With repeated bending, some of the optical fibres within the bundle will break** and light may escape through the cracks. This degrades the signal transmission through the fibres.

Google Forms Microsoft Forms

Quick check 11

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Only black and white images can be observed when endoscopes are used. (T/F)

2 *True or false:* Tissue sample can be obtained by using the instruments passed through the endoscope. (T/F)

Exercise 1.3

1 Which of the following organs can be inspected by using an endoscope?

A Heart
B Liver
C Brain
D Stomach

2 The refractive index of the core of an optical fibre is 1.75 and the core is protected by cladding. The refractive index of the cladding must be

A smaller than 1.75.
B larger than 1.75.
C close to 1.75.
D differing greatly from 1.75.

anaesthesia 麻醉

*3 A letter 'L' is observed through two bundles of optical fibres (Fig a). One is a coherent bundle and the other is an incoherent bundle. Each bundle contains a large number of optical fibres.

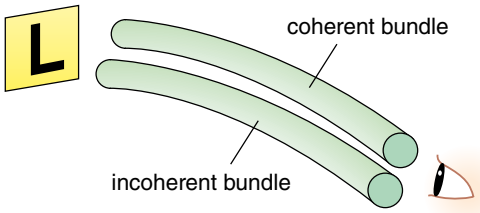
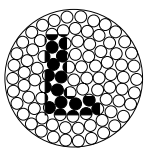
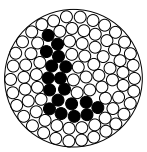
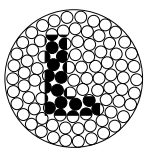
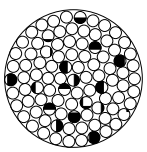
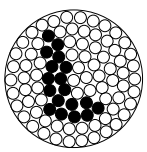
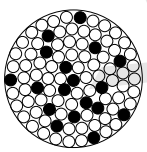
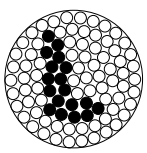
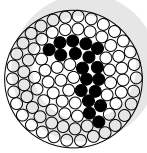


Fig a

Which of the following best represents the images as observed through the two bundles?

	Coherent bundle	Incoherent bundle
A		
B		
C		
D		

*4 Which of the following statements about a fibre optic endoscope is **incorrect**?

A It only provides black and white images of the inspected organ.

B It provides a real-time view of the inspected organ.

C Tools can be inserted through the duct in the endoscope to obtain tissue for further tests.

D It uses visible light for transmitting images.

*5 Which of the following statements about coherent and incoherent bundles fibres is/are correct?

(1) Both can transmit light.

(2) Both can transmit an image.

(3) Both can function normally when bent.

A (3) only B (1) and (2) only

C (1) and (3) only D (1), (2) and (3)

6 Describe how the coherent bundle fibres can transmit images from one end to another.

7 Should the cladding of an optical fibre have a smaller or larger refractive index than the core? Explain your answer.

8 State two advantages of using endoscopes in diagnosis.

9 State two limitations of using endoscopes in diagnosis.

10 State the function of each of the following components in an endoscope.

(a) Optical fibres

(b) Objective lens

(c) Eyepiece

*11 (a) If an optical fibre has no cladding, would it be easier for light to get transmitted along the fibre? Explain briefly.

(b) State a reason why the core is covered by a cladding.

*12 The bending section of an endoscope is inserted into a patient's stomach. Describe how the image of the inner surface of the stomach is formed and observed by the doctor.

*13 The optical fibres in an endoscope usually consist of an inner core with cladding (Fig b). The cladding and the core are made of two different materials *A* and *B*.

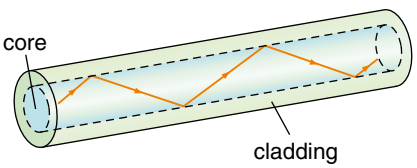


Fig b

(a) *A* and *B* have refractive indices 1.45 and 1.70 respectively. Which material should be used to make the cladding? Explain your answer.

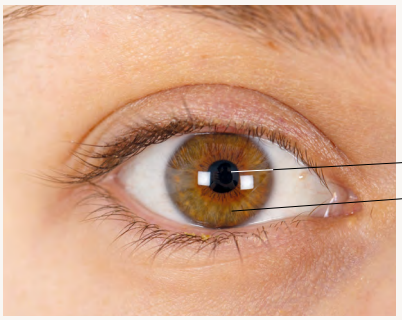
(b) Find the critical angle at the core-cladding interface.

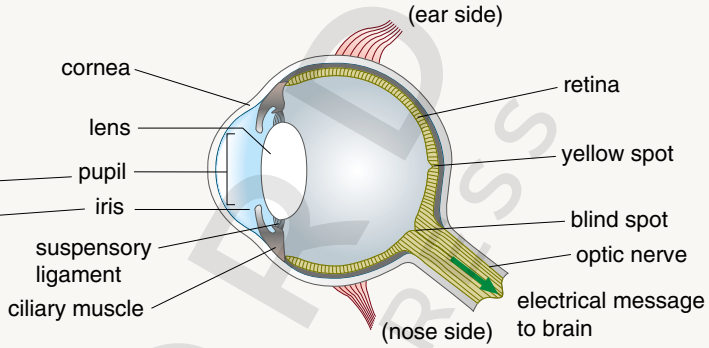
(c) State the uses of incoherent bundle fibres and coherent bundle fibres in a fibre optic endoscope.

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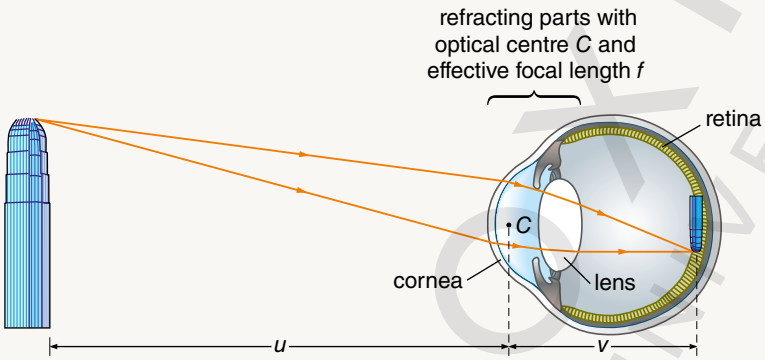
Key concepts 1

1 Structure of the human eye





2 Accommodation



Accommodation

adjust thickness of lens (focal length) so that the image of an object falls on the retina

Near object ($u\downarrow$)	Far object ($u\uparrow$)
• ciliary muscles contract • lens thicker ($f\downarrow$) • more converging	• ciliary muscles relax • lens thinner ($f\uparrow$) • less converging

Related formulae

Lens formula: $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

Power: $P = \frac{1}{f}$

Effective power of several lenses = sum of powers of all lenses

Near point and far point

Near point

- closest point to the eye at which an object can be placed and still form a sharp image on the retina
- at about 25 cm for a normal eye

Far point

- furthest point from the eye at which an object can be placed and still form a sharp image on the retina
- at infinity for a normal eye

3 Light sensitive cells

Light sensitive cells act as transducers to convert light energy into electrical signals.



Rods

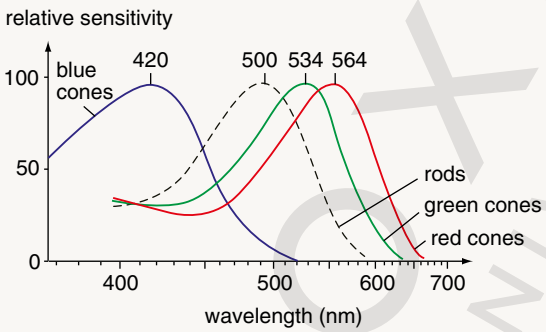
- provide black and white vision
- provide vision in low light condition



Cones

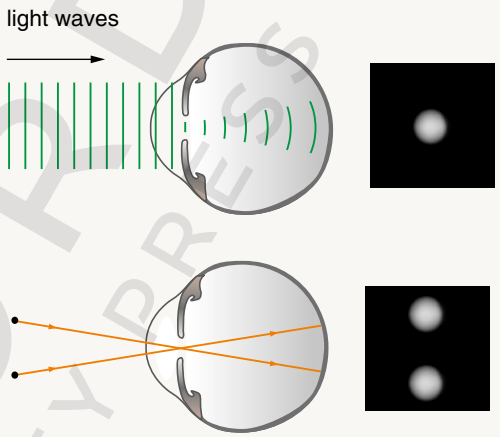
- provide colour vision
- give fine detail for objects directly in front of the eye

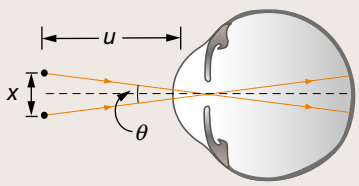
Receptor absorption curves



4 Resolving power

Since light is diffracted when travelling through the pupil, the images of two points may overlap if they are too close.





- Angular separation: $\theta \approx \frac{x}{u}$
- Minimum angular separation of two resolvable points: $\theta_R \approx \frac{1.22\lambda}{d}$
- Two points can be resolved if $\theta > \theta_R$

5 Defects of vision



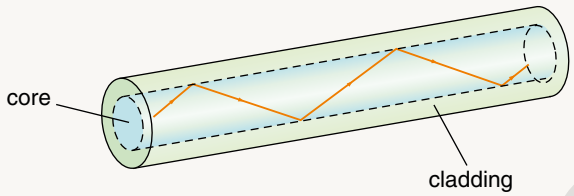
	Short-sightedness (Myopia)	Long-sightedness (Hypermetropia)	Old-sightedness (Presbyopia)
Problem	• far point closer to eye than ∞ • cannot see distant objects clearly	• near point farther from eye than 25 cm • cannot see near objects clearly	• near point farther from eye than 25 cm • cannot see near objects clearly
Cause	eyeball too long or lens too thick	eyeball too short or lens too thin	the elasticity of lens decreases with age
Correction	diverging lens	converging lens	converging lens

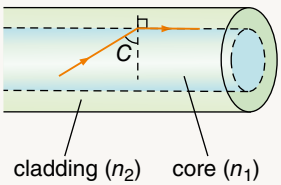
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6 Optical fibres and fibre bundles

Optical fibre

- a core surrounded by cladding
- cladding has smaller refractive index
- light travels along through a series of total internal reflections



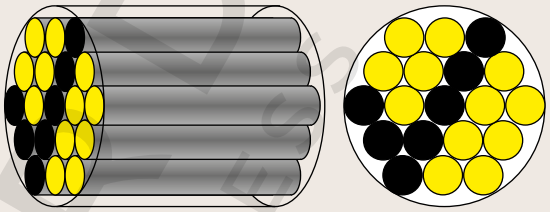


Critical angle

$n_1 \sin C = n_2 \sin 90°$

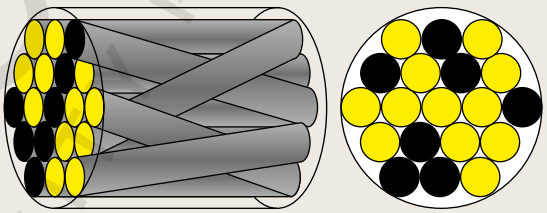
Coherent bundle fibre

relative position of each fibre is preserved from one end to the other

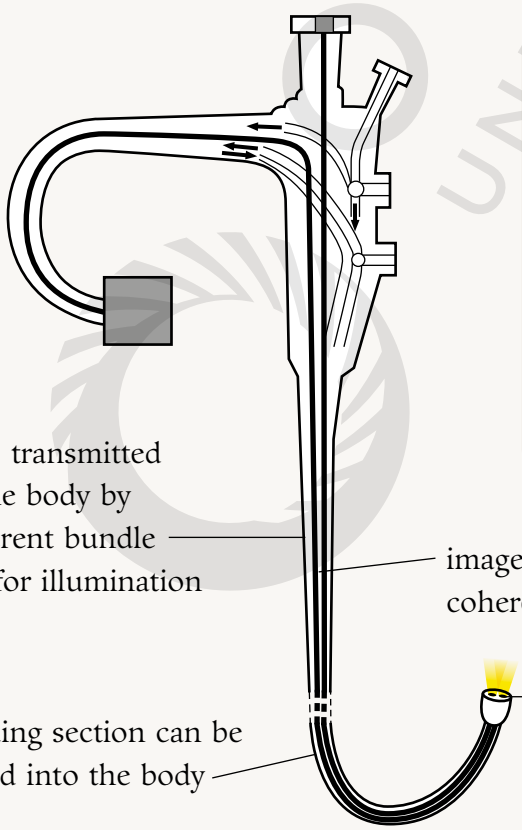


Incoherent bundle fibre

relative position of fibres are not preserved



7 Fibre optic endoscopy



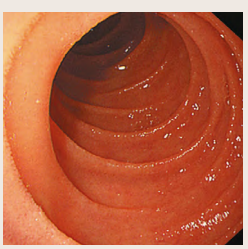
light is transmitted into the body by incoherent bundle fibres for illumination

image is transmitted out of the body by coherent bundle fibres for observation

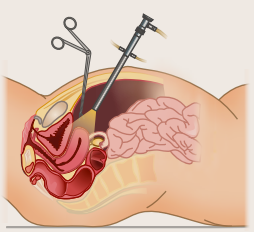
small instruments can be passed to the tip to obtain sample or perform simple operations

a bending section can be inserted into the body

Common uses of endoscopes



observe inner surface of hollow organs



perform minimally invasive surgery

Key terms

1 accommodation 視覺調節 p.7
2 angular separation 角間距 p.16
3 cladding 包覆層 p.39
4 coherent bundle fibre 相干光纖管束 p.42
5 cone cell 視錐細胞 p.11
6 dioptre 屈光度 p.22
7 far point 遠點 p.23
8 fibre optic endoscope 光纖內窺鏡 p.38
9 fibre optic endoscopy 光纖內窺鏡檢查術 p.38
10 incoherent bundle fibre 不相干光纖管束 p.43
11 long-sightedness / hypermetropia 遠視 p.27
12 near point 近點 p.23
13 old-sightedness / presbyopia 老花 p.28
14 power 焦強 p.22
15 receptor absorption curve 感受器吸收曲線 p.13
16 resolving power 解像能力 p.16
17 rod cell 視桿細胞 p.11
18 short-sightedness / myopia 近視 p.26
19 transducer 換能器 p.11

Revision 1

Multiple-choice questions

★1 1.2 Chris is suffering from short-sightedness and his uncorrected far point is at 2 m from the eye. Find the power of the spectacles required to correct the defect.

A −0.5 D
B +0.5 D
C −2.0 D
D +2.0 D

★2 1.2 Mandy is suffering from an eye defect. Figure a is a ray diagram showing how she sees a distant object when she is wearing a pair of corrective spectacles.

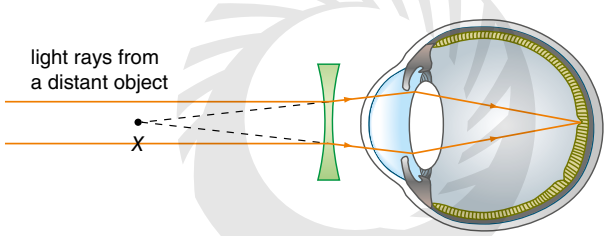


Fig a

What is position *X*?

A The corrected near point
B The corrected far point
C The uncorrected near point
D The uncorrected far point

★3 1.2

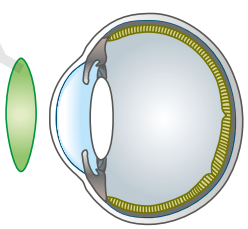


Fig b

When a person suffers from long-sightedness, he can wear a corrective lens (Fig b) so that

A the near point is brought nearer.
B the near point is brought further away.
C the far point is brought nearer.
D the far point is brought further away.

★4 1.3 Which of the following statements about fibre optic endoscopy is/are correct?

(1) All parts of the human body can be inspected with this method.
(2) Instead of coherent bundle fibres, incoherent bundle fibres are usually used to transmit light for illumination because the cost is lower.
(3) The cladding of the optical fibre prevents the loss of signal during transmission.

A (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

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*5 1.1 The graph shows the changes in thickness of the lens of a person's eye in a period of 5 seconds (Fig c).

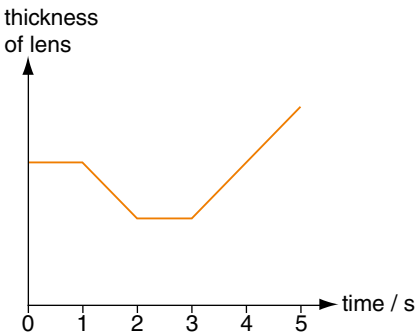


Fig c

Which of the following is the time period when the person is looking at an object moving away from him?

A From 0–1 s
B From 1–2 s
C From 2–3 s
D From 3–5 s

*6 1.2 Which of the following statements about a lens with a power of –10 D is/are correct?

(1) It is a concave lens.
(2) Its focal length is 10 cm.
(3) It bends light more than a lens with a power of –5 D.

A (2) only
B (1) and (2) only
C (1) and (3) only
D (1), (2) and (3)

*7 1.2 The far point of an eye is at infinity. If the powers of the eye are +41 D and +38 D when looking at an object at its near point and far point respectively, how far is the near point from the eye?

A 25 cm
B 28.5 cm
C 30.3 cm
D 33.3 cm

*8 1.2 Betty is wearing concave spectacles with a power of –1.25 D. The range of accommodation of her eye without the spectacles is from 25 cm to 80 cm. Find the range of accommodation of her eye with the spectacles.

A From 28.3 cm to 80 cm
B From 28.3 cm to infinity
C From 36.4 cm to 80 cm
D From 36.4 cm to infinity

**9 1.1

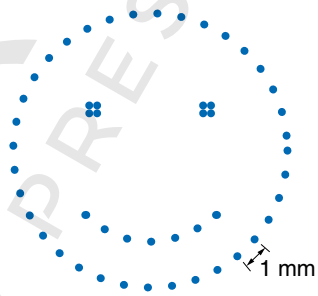


Fig d

Figure d shows a smiley face formed by small blue dots reflecting light of wavelength 450 nm. The distance between adjacent blue dots forming the outline of the face is 1 mm. Jack looks at the picture from a distance d. He sees the face without realizing that its outline is formed by small blue dots. Given that the diameter of his pupil is 2 mm, find the minimum value of d.

A 1 mm B 2 mm
C 1.82 m D 3.64 m

Refer p.16, 17

10 HKDSE 2013 Paper 2 Q4.3 1.3

The diagram below represents two coherent optical fibre bundles X and Y used in endoscopes. Their cross-sections have the same dimensions but X has more and finer fibres. Which statements are correct?

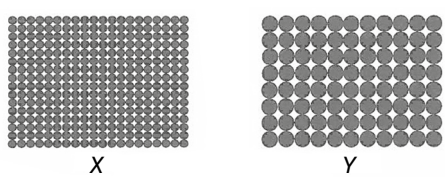


Fig e

(1) X gives a much brighter image than Y.
(2) X can be bent more than Y.
(3) X gives an image of higher resolution than Y.

A (1) and (2) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)
(38%)

11 HKDSE 2014 Paper 2 Q4.1
1.2

Mr Lee wears a pair of bifocal lenses as shown in Figure f. The respective powers of the upper half and the lower half of each lens are −1.5 D and +2.0 D. Which of the following statements is/are correct?

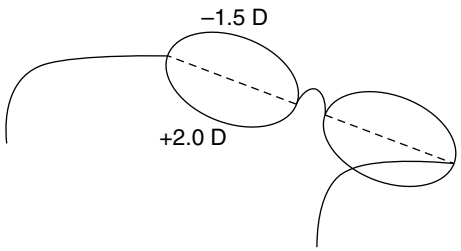


Fig f

(1) The upper half is for viewing distant objects while the lower half is for viewing objects at a close distance.

(2) Mr Lee only suffers from old sight (presbyopia).

(3) Without spectacles, Mr Lee cannot see an object clearly no matter how far it is placed from him.

A (1) only (48%)

B (3) only

C (1) and (2) only

D (2) and (3) only

12 HKDSE 2016 Paper 2 Q4.1
1.1

The retina contains two types of light sensitive cells, namely, rods and cones. Which of the following statements about rods and cones is/are correct?

(1) Rods are responsible for vision at low light levels.

(2) Rods are less numerous than cones.

(3) Both rods and cones are capable of colour vision.

A (1) only (48%)

B (1) and (2) only

C (2) and (3) only

D (1), (2) and (3)

13 HKDSE 2018 Paper 2 Q4.1
Misc

The refractive indices of air and the components of the eye are listed below. Which pairs of media provide the most bending of light due to refraction?

medium	refractive index
air	1.00
cornea	1.38
aqueous humour	1.34
surface of lens	1.39
centre of lens	1.41

A surface of lens – centre of lens

B aqueous humour – surface of lens

C cornea – aqueous humour

D air – cornea (64%)

14 HKDSE 2018 Paper 2 Q4.5
1.3

As shown below the optical fibre used for an endoscope is cladded with material of refractive index different from that of the fibre core to avoid light leakage. The maximum angle of incidence i_{max} is 38.4° for no leakage of light. Find the refractive index of the cladding material.

Given: refractive index of the fibre core = 1.55

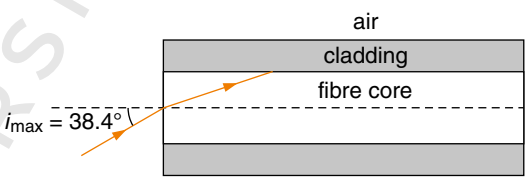


Fig g

A 1.69 B 1.42 (54%)

C 1.33 D 1.16

15 HKDSE 2019 Paper 2 Q4.1
1.2

John suffers from long-sightedness. After wearing suitable corrective spectacles, how would his near-point distance and far-point distance be affected?

	near-point distance	far-point distance
A	increased	increased
B	increased	unchanged
C (20%)	decreased	decreased
D	decreased	unchanged

16 HKDSE 2019 Paper 2 Q4.2
1.2

An object is placed 20 cm in front of a concave lens. The magnification of the image is 0.5. Find the power of the lens.

A +20 D B −5 D (42%)

C −10 D D −20 D

Conventional questions

*17 1.1 Three small LEDs are arranged as shown in Figure h. Brian stands at a distance where the three LEDs cannot be resolved by his eyes.

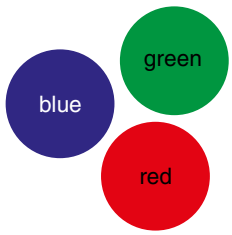


Fig h

(a) What colour is perceived if all the lights are switched on? (1 mark)

(b) Explain why he can see one colour instead of three colours. (2 marks)

*18 1.1 Karen is looking at a card board with red light passing through the slits on it. Each slit is separated by 1 mm (Fig i). The wavelength of red light is 680 nm. The diameter of the pupil of Karen's eye is 6 mm.

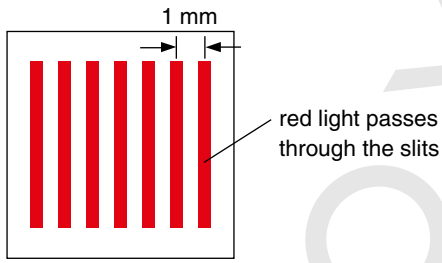


Fig i

(a) Estimate the minimum angular separation of two slits so that they can be resolved by Karen's eye. (2 marks)

(b) How far should the card board be placed from Karen so that she can just distinguish each slit? (2 marks)

(c) Assume the card board is placed as the distance in (b). Karen thinks that she can still distinguish each slit if the red light is replaced by a violet light. Is she correct? Explain briefly. (3 marks)

*19 1.2 Tom is suffering from long-sightedness. The power of his eye is +40 D when looking at an object at infinity.

(a) Find the distance between the retina and the optical centre of the refracting parts of Tom's eye. (2 marks)

(b) Tom wears a pair of convex lenses with a power of +3 D and the near-point distance of his eye with the lenses is at 25 cm.

(i) Does a high-power lens or low-power lens bend light more? (1 mark)

(ii) Find the effective power of the combination of his eye and the convex lenses when he is looking at an object at 25 cm. (2 marks)

(iii) Find the near-point distance of Tom's eye without the lenses. (2 marks)

(c) Both long-sightedness and old-sightedness lead to problems with near vision. Compare the causes of these two defects of vision. (2 marks)

*20 1.2 Ada is suffering from a defect of vision and is wearing a pair of corrective lenses with a power of –2.5 D.

(a) What kind of defect of vision is Ada suffering from? State a possible cause of this defect. (2 marks)

(b) (i) Describe the problem in vision that Ada has without the corrective lenses. (1 mark)

(ii) Draw a ray diagram to illustrate the problem in (i). (2 marks)

(c) What type of lens do the corrective lenses belong to? (1 mark)

(d) Consider an object at infinity. *I* is its image formed by the corrective lens. Find the distance of *I* from the lens. State the physical meaning of this image position. (3 marks)

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***21** 1.3 An endoscope uses a coherent bundle to transmit images and an incoherent bundle to transmit light for illumination.

(a) With the aid of a diagram, describe what a coherent bundle is and explain how it transmits images. (3 marks)

(b) Tommy thinks that the endoscope can still function normally if the incoherent bundle is also replaced by another coherent bundle. Is he correct? Explain briefly. (2 marks)

(c) It is known that the critical angle at the core-cladding interface of an optical fibre is 65°. Find the ratio of the refractive index of the core to that of the cladding. (2 marks)

***22** 1.2 As Sherry gets older, she can only focus on objects at a distance between 32 cm to 1.2 m.

(a) (i) Define the term 'near point'. (1 mark)

(ii) Where is the near point of Sherry's eye? (1 mark)

(b) In Figures j and k, complete the paths of the light rays to illustrate how Sherry looks at a distant object and a near object. (2 marks)

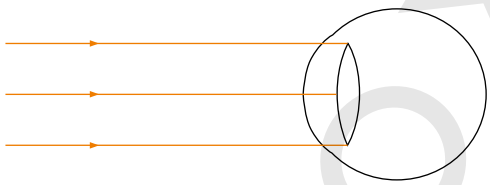


Fig j

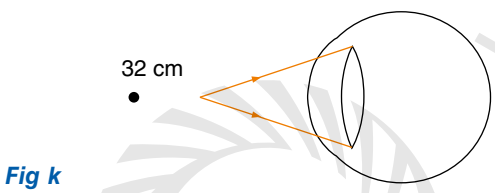


Fig k

(c) Sherry now wears spectacles of power +0.875 D.

(i) Can she see an object 20 cm from her eye clearly? Show your calculation. (3 marks)

(ii) Draw a ray diagram to illustrate the situation in (i). Indicate the corrected near point in the diagram. (3 marks)

***23** 1.1 Figure l shows the receptor absorption curves of rods and green cones.

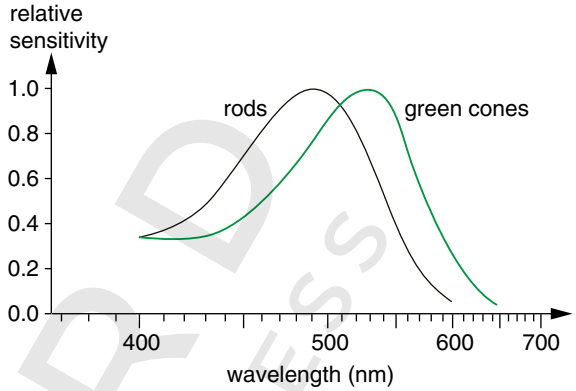


Fig l

(a) How are the rods and cones distributed on the retina? (2 marks)

(b) The wavelength of green light is about 530 nm. Explain why it is easier to find an emergency exit sign in the dark if it is green in colour (Fig m). (1 mark)

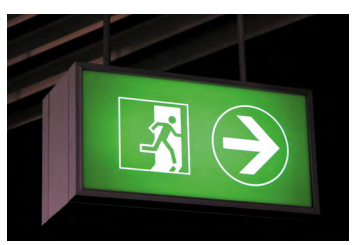


Fig m

***24** 1.1 Eric is walking in the park at night and is looking at the Moon.

(a) In Figure n, draw a ray diagram to show the paths of light rays entering Eric's eye when he is looking at a point on the Moon. (2 marks)

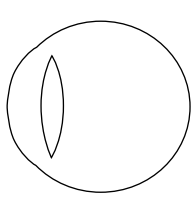


Fig n

(b) After looking at the Moon, he turns to look at the objects in the park. Describe the changes that occur in his eye. (2 marks)

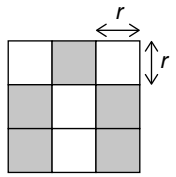
(c) Explain why Eric cannot distinguish the colour of these objects in such dim light conditions. (2 marks)

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25 **HKDSE 2014 Paper 2 Q.4(b)**
Misc

(a) John has normal eyesight and the power of his eye is +59 D in viewing distant objects. Estimate the separation between the lens and the retina of his eye. Assume that the refracting power is mainly contributed by the eye lens. (2 marks)

(b) The display panel of a smart phone *X* is made up of numerous tiny square pixels as shown (Fig o).



square pixels of part of the display panel

Fig o

John is looking at the graphics on the display panel of smart phone *X*. The diameter of his eye's pupil is 4.0 mm. Estimate the resolving power θ (in radians) of his eye for graphics in green colour.

Given: wavelength of green light $= 5.35 \times 10^{-7}$ m (2 marks)

(c) The pixels of smart phone *X* are so small that the human eye is unable to distinguish two adjacent pixels at a typical viewing distance $L = 0.30$ m. Using the result of (b), estimate the maximum length of a side of a square pixel, *r*, on the display panel of smart phone *X*. You may assume that for small angle θ in radians, $\tan\theta = \theta$. (2 marks)

Q25 Exam Report: Parts (a) and (b) were well answered. However, some had difficulties in manipulating radian and made wrong or unnecessary conversion of angles. Part (c) was poorly answered. Many candidates had no idea about the approximation $\tan\theta \approx \theta$ when θ is small. Not many were able to apply the formula of arc length ($s = r\theta$) correctly, which should be $r = L\theta$ here. Some confused *r* in the first formula and the *r* in the question and used it in place of *L*. Some candidates wrongly used *r*/2 instead of *r*.

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Know more

The ciliary muscles and the suspensory ligaments (p.8)

When an eye is looking at a near object, the ciliary muscles contract, reducing the tension in the suspensory ligaments. The pull on the lens is then reduced, allowing the lens to be thickened (Fig a). When looking at a distant object, the ciliary muscles relax, increasing the tension in the suspensory ligaments. The lens is then pulled thin (Fig b).

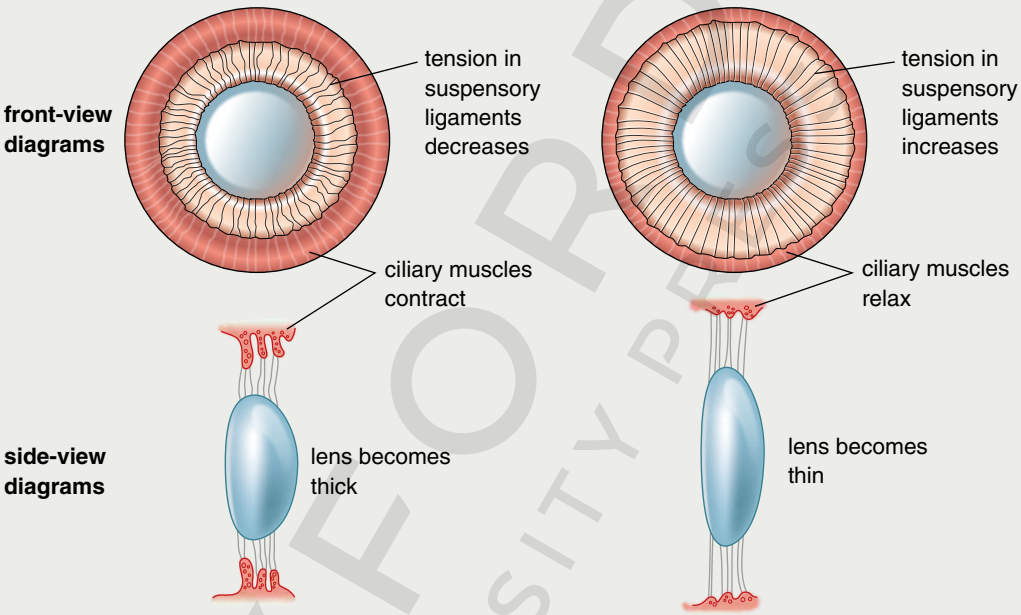


Fig a *Looking at a near object.*

Fig b *Looking at a distant object.*

Seeing objects at different distances at the same time (p.8)

It is not possible to see a near object and a far object clearly at the same time. When light from either one converges on the retina, light from the other will be spread over an area on the retina (Fig c), making it look blurred.

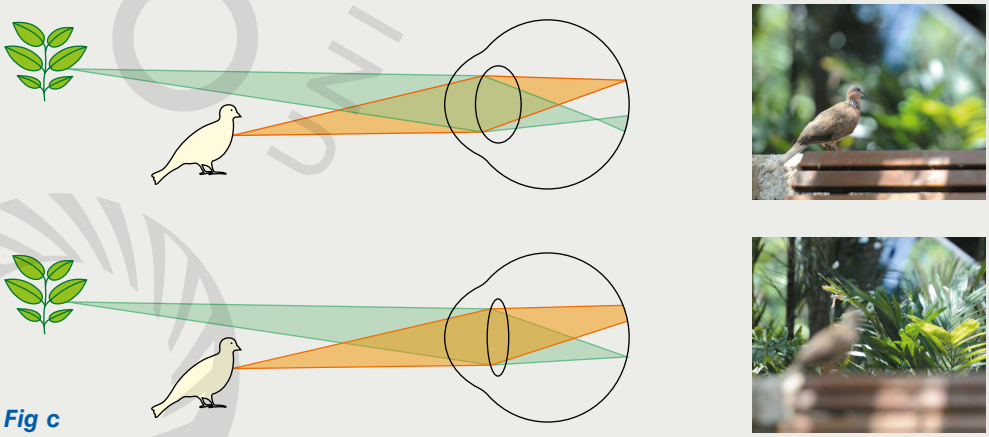


Fig c

However, this can be improved by looking through a small hole. In this way, the light entering the eye only passes through part of the lens and the light will be spread over a much smaller area on the retina (Fig d), making it look less blurred.

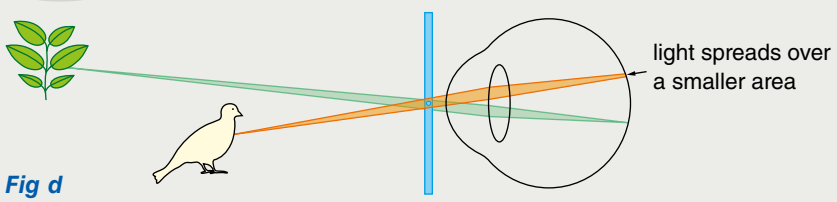


Fig d

✎ The focal length of the lens in low-end cameras cannot be changed much. Such cameras achieve sharp images for both near and distant objects by using a small opening (aperture). Our eyes are more like high-end cameras!

Know more

Structure of a fibre optic endoscope (p.43)

A fibre optic endoscope can be divided into three major parts: a handle, a bending section and an endoscope support system.

